

On the Universality of Comparability Grids for Measurement-Based Quantum Computation

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Abstract

The measurement-based quantum computing (MBQC) model presents a way to do quantum computations by measuring a pre-prepared graph state. As such, there is interest in determining which graph states are universal for MBQC. The universality of the comparability grid is currently unknown and of some interest, since if it is universal, so are all families of graph states with unbounded rank-width. In this thesis, we present several approaches toward proving this universality, culminating in a construction conjectured to enable universal quantum computation on the comparability grid through MBQC.

Acknowledgements

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I would also like to extend my gratitude to Du (Dan) Van, the primary collaborator of this project, who has always been available to listen to my ideas. I wish him the best of luck in his future studies at NUS. I would also like to thank Jackson Morris, who has always been available to mentor and provide feedback for this project. I am also thankful to Kunal Marwaha, for helpful discussions and for sharing his notes on graph states [Mar].

I also thank my family and friends for their constant support in all aspects of my academic journey. It is because of them that I may focus on my studies and pursue my aspirations without distraction.

Author's Note

One of the great difficulties I faced writing this thesis was figuring out where to stop. In other words, what may I assume of the reader, and what must be explained? Past Mathematics undergraduate honors theses have been most helpful in this regard. Many of these take an introductory stance, writing for a mathematically mature but non-specialized audience (i.e. a senior mathematics undergraduate).

For this reason, I have prepared a deliberately extensive exposition. For example, the ZX-Calculus, Clifford circuits, and several other concepts reviewed in this thesis are well-known to the quantum computing community. Nevertheless, many modern publications in this area review these concepts in their introductions [Hah+26; Bac+21; MB23].

Finally, much of the research connected to this thesis remains ongoing. Rather than presenting a finished story, this thesis aims to record a path through the problem starting from first principles. I hope the reader may also find value and enjoyment in this path.

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1 Introduction

We start with an informal introduction to quantum computing and motivate the central problem.

Fault-tolerant quantum computing has the potential, through algorithms like Shor’s factoring algorithm [Sho97], to deliver exponential speedups for certain problems over the best known classical algorithms. However, current Noisy Intermediate-Scale Quantum (NISQ) computers have too few reliable qubits and too much noise to deliver practical benefit [Pre18].

Quantum computing admits a variety of computational models: The quantum Turing machine [Deu85], the quantum circuit model [Bar+95], adiabatic quantum computation [Far+01], and measurement-based models [RB01], among many others. Though these models are equivalent from a complexity viewpoint (informally, any problem that is efficiently solvable in one is efficiently solvable in the others) [RB01; Aha+08; Chi93], the underlying quantum properties used by each may differ significantly.

The measurement-based quantum computation (MBQC) model is of particular interest. This is because MBQC conducts quantum computation by adaptively measuring an entangled resource state. This is quite shocking, as it means that any quantum computation may be done in constant depth. The cost of this model, however, is that measurements need to be adaptive. In other words, processing must be done on-the-fly to determine what qubits should be measured next and how.

Since measurement is irreversible, MBQC leads to a model of one-way quantum computation. In other words, all required entanglement must be initially present in the resource state. Informally then, entanglement is the resource consumed by this model of quantum computation. Thus, there is interest in determining what resource states are universal for quantum computation.

Graph states are a kind of resource state whose data corresponds to a simple, undirected graph. The comparability grid is a graph with vertices $\{(x, y) \mid x, y \in [n]\}$ and an edge between (x_1, y_1) and (x_2, y_2) if $x_1 \geq x_2$ and $y_1 \geq y_2$ or $x_2 \geq x_1$ and $y_2 \geq y_1$, excluding self-loops. It is an open problem whether the family of all comparability grids is universal for MBQC. There are implications for both possibilities. For example, if comparability grids are not universal, then unbounded rank-width is not a sufficient condition for the universality of a family of graph states, since the family of all comparability grids has unbounded rank-width.

In this thesis, we report some progress on this problem. Concretely, we give a set of constructions for how one may be able to extract universal quantum computation using comparability grid graph states as our resource state. In tandem with this, we outline various algorithms to verify small cases of these conjectures.

Readers with no prior exposure to quantum computation are encouraged to start at Appendix A. In Section 2, we introduce the notation and background so that in Section 3 we may formally define the concepts in our problem statement. Then, Section 4 recalls some helpful results regarding comparability grids. These theoretical results guide the algorithmic approaches in Section 5, which provide evidence for the conjectures in Section 6.

2 Background

First some notation. Let $\mathbb{N} = \{1, 2, \dots\}$. For $k \in \mathbb{N}$, define $[k] := \{1, \dots, k\}$ and $[0] := \emptyset$. Let I be the identity matrix of any size and of any vector space, which will be clear from context. For a graph G , let $V(G)$ denote its vertex set and $E(G)$ its edge set. For any vector v with n entries, let v_i be its i th entry for all $i \in [n]$. For any matrix A with n rows and m columns, let $A_{i,j}$ be the entry at the i th row and j th column, for all $i \in [n], j \in [m]$.

We assume the reader is familiar with the mathematical foundations of quantum computing e.g. at the level of an upper undergraduate quantum computing class. Please see Appendix A otherwise.

2.1 The ZX-Calculus

DEFINITION 2.1 The **ZX-Calculus** is a graphical language for representing quantum computation. Objects of the ZX-Calculus, called **ZX Diagrams**, are made up of two types of generators known as **spiders**: The **X-spider**, with n inputs and m outputs, is defined

$$n \text{ --- } \textcircled{\alpha} \text{ --- } m := |+\rangle^{\otimes m} \langle +|^{\otimes n} + e^{i\alpha} |-\rangle^{\otimes m} \langle -|^{\otimes n}. \quad (2.1)$$

We say that the above spider has a **phase** $\alpha \in [0, 2\pi)$. Similarly, the **Z-spider** is defined

$$n \text{ --- } \textcircled{\alpha} \text{ --- } m := |0\rangle^{\otimes m} \langle 0|^{\otimes n} + e^{i\alpha} |1\rangle^{\otimes m} \langle 1|^{\otimes n}. \quad (2.2)$$

In other words, the ZX-Calculus is a calculus (i.e. rules) for manipulating Z and X spiders.

REMARK 2.2 In the literature, the X-spider is often red and the Z-spider is often green, e.g. ● and ●. However, for accessibility reasons we choose to adopt the black-and-white ZX-Calculus color scheme instead.

DEFINITION 2.3 Write $a \propto b$ if $a = c \cdot b$, for some $0 \neq c \in \mathbb{C}$. Furthermore, if D is a ZX diagram, write $\llbracket D \rrbracket$ to denote the matrix represented by D . To reduce notational clutter, we will use $\llbracket D \rrbracket$ only when it is necessary to distinguish a diagram from its matrix.

EXAMPLE 2.4 The standard quantum states can be written as spiders without inputs:

$$\textcircled{0} \text{ --- } = \sqrt{2} |0\rangle \propto |0\rangle, \quad \textcircled{\pi} \text{ --- } = \sqrt{2} |1\rangle \propto |1\rangle, \quad (2.3)$$

$$\textcircled{0} \text{ --- } = \sqrt{2} |+\rangle \propto |+\rangle, \quad \textcircled{\pi} \text{ --- } = \sqrt{2} |-\rangle \propto |-\rangle. \quad (2.4)$$

Single qubit gates can be written as spiders with one input and one output:

$$\text{---} \textcircled{\alpha} \text{ ---} = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\alpha} \end{bmatrix} = e^{i\alpha/2} \begin{bmatrix} e^{-i\alpha/2} & 0 \\ 0 & e^{i\alpha/2} \end{bmatrix} = e^{i\alpha/2} R_Z(\alpha), \quad (2.5)$$

$$\text{---} \textcircled{\alpha} \text{ ---} = \frac{1}{2} \begin{bmatrix} 1 + e^{i\alpha} & 1 - e^{i\alpha} \\ 1 - e^{i\alpha} & 1 + e^{i\alpha} \end{bmatrix} = e^{i\alpha/2} R_X(\alpha). \quad (2.6)$$

Thus $\text{---} \boxed{Z} \text{---} \propto \text{---} \textcircled{\pi} \text{---}$ and $\text{---} \boxed{X} \text{---} \propto \text{---} \textcircled{\pi} \text{---}$.

DEFINITION 2.5 Informally, the **composition** of spiders in ZX diagrams is similar to the composition of quantum gates in quantum circuits: Spiders connected via wires matrix multiply, while spiders “stacked vertically” are tensored together.¹

DEFINITION 2.6 Here is an example of ZX composition: The Hadamard gate satisfies

$$\boxed{H} = e^{-i\pi/4} \text{---} \textcircled{\pi/2} \text{---} \textcircled{\pi/2} \text{---} \textcircled{\pi/2} \text{---}. \quad (2.7)$$

Due to this verbose representation, we shorthand $\text{---} \square \text{---} := \boxed{H}$ and call this the **H-box**.

DEFINITION 2.7 **Wires** or lines between spiders may “bend” in the following ways:

$$\text{X} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \text{C} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad \text{D} = \begin{bmatrix} 1 & 0 & 0 & 1 \end{bmatrix}. \quad (2.8)$$

These are called “swap,” “cup,” and “cap,” respectively from left to right, and their composition is done by matrix multiplication.

DEFINITION 2.8 The **arity** of a spider is the number of inputs and outputs it has. If a spider has phase 0, we say it is **phase-less**, and often do not bother writing the 0. Namely, $\bigcirc := \textcircled{0}$ and $\bullet := \textcircled{0}$.

DEFINITION 2.9 The ZX Calculus admits a finite set of **rewrite rules**, or graphical equivalences for rewriting ZX diagrams. These are summarized in Table 1. For further explanation of these rules and some history on their names, please see [Wet20].

So far, it is not clear why we would choose to use ZX-Calculus over either the algebraic or circuit representation of a quantum circuit. This becomes clearer when we look at Table 1. Namely, Table 1 tells us that there is a short, finite set of rules for rewriting any ZX diagram to another. Paired with the fact that there are only two types (three if you count the H-box) of objects being manipulated, this notation allows us to reason about quantum computation at a higher level compared to the matrix or circuit representations. The following lemma is an example of this.

LEMMA 2.10 (COLOR CHANGE LEMMA) Any rewrite identity in ZX holds if we replace Z-spiders by X-spiders and vice versa.

¹Formally, the category of ZX diagrams, $\mathbf{ZX}$, is a compact closed symmetric monoidal category. The objects are non-negative integers and the morphisms ZX diagrams. The composition of morphisms correspond to the composition of ZX diagrams. The tensor product on objects satisfies $n \otimes m := n + m$ and the tensor product of two ZX diagrams is given by “stacking” the diagrams on top of another. Due to the existence of the cup and cap, this is a compact closed category. Interested readers should see [AC07] for more details.

Name	Rewrite rule	Description
Spider fusion (f)	$ \begin{array}{c} n_1 \vdots \alpha \\ \vdots \\ n_2 \vdots \beta \end{array} \begin{array}{c} m_1 \\ \vdots \\ m_2 \end{array} = \begin{array}{c} n_1 \\ \vdots \\ n_2 \end{array} \begin{array}{c} \alpha+\beta \\ \vdots \\ \alpha+\beta \end{array} \begin{array}{c} m_1 \\ \vdots \\ m_2 \end{array} $	Spiders of the same color connected by any number of wires fuse and their phases add.
Identity removal (id)	$ \text{---} \bigcirc \text{---} = \text{---} $	Spiders with one input, one output, and phase 0 are the identity matrix and can be removed.
Hadamard cancellation (hh)	$ \text{---} \square \square \text{---} = \text{---} $	Two connected Hadamards cancel each other.
π -commutation (π)	$ \text{---} \pi \text{---} \alpha \text{---} = e^{i\alpha} \text{---} -\alpha \text{---} \pi \text{---} $	A π phase copies through a spider of the opposite color and flips its phase.
State copy (cp)	$ \text{---} a\pi \text{---} \alpha \text{---} = e^{ia\alpha} \sqrt{2}^{-(n-1)} \text{---} a\pi \text{---} \pi \text{---} $	A π phase copies through a spider of the opposite color and flips its phase.
Color change (cc)	$ \begin{array}{c} \vdots \\ \vdots \end{array} \alpha \begin{array}{c} \vdots \\ \vdots \end{array} = \begin{array}{c} \vdots \\ \vdots \end{array} -\alpha \begin{array}{c} \vdots \\ \vdots \end{array} $	Commuting Hadamards through a spider changes the color.
Bi-algebra (b)	$ \begin{array}{c} \vdots \\ \vdots \end{array} \text{---} \text{---} \begin{array}{c} \vdots \\ \vdots \end{array} = \sqrt{2}^{(n-1)(m-1)} \begin{array}{c} \vdots \\ \vdots \end{array} \begin{array}{c} \vdots \\ \vdots \end{array} $	An adjacent phase 0 X and Z spider can commute past one another to create a complete bipartite graph with sides of the opposite color.
Hopf rule (hf)	$ \text{---} \bigcirc \bigcirc \text{---} = \frac{1}{2} \text{---} \bigcirc \text{---} \bigcirc \text{---} $	If phase 0 spiders of opposite color are connected by multiple wires, we may remove the wires pairwise.

Table 1: ZX-Calculus Rewrite Rules

Proof. Consider a ZX diagram identity $D = D'$. For both: Apply (cc) on all spiders to swap all colors. All inputs and outputs connected to a spider gain a H-box, which are the same wires in D and D' . All wires between two spiders gain two H-boxes, which by (hh) cancel. Thus the color-swapped D is equivalent to the color-swapped D' . $\square$

THEOREM 2.11 Any ZX diagram is topologically invariant; only connectivity matters.

Proof. The proof surmounts to a tedious enumeration of all possible rearrangements a quantum circuit [CK17]. Like in quantum circuits, we often write ZX diagrams where inputs come from the left and outputs go to the right. This is simply convention. More formally, it suffices to maintain an ordered list of wires considered input(s) and output(s). $\square$

EXAMPLE 2.12 The CNOT gate has the following ZX representation:

$$\begin{array}{c} \bullet \\ | \\ \oplus \end{array} = \begin{array}{c} \circ \\ | \\ \ominus \end{array}. \quad (2.9)$$

What Theorem 2.11 says is that

$$\left[\begin{array}{c} \circ \\ | \\ \ominus \end{array} \right] = \left[\begin{array}{c} \circ \\ | \\ \bullet \end{array} \right] = \left[\begin{array}{c} \bullet \\ | \\ \oplus \end{array} \right]. \quad (2.10)$$

LEMMA 2.13 The ZX-Calculus is exactly universal for quantum computation², up to some complex scalar factor.

Proof. By Example 2.4 and Euler decomposition (Theorem A.12), we have that every single-qubit gate can be represented in the ZX-Calculus, up to some complex scalar factor. Example 2.12 gives us a representation of CNOT, a two-qubit entangling gate in the ZX-Calculus. By Theorem 1.3 of [BB01], arbitrary single-qubit gates with any entangling 2-qubit gate are exactly universal for quantum computation. $\square$

THEOREM 2.14 Any $c \in \mathbb{C}$ can be written as a ZX diagram.

Proof. Notice³

$$\left[\begin{array}{c} \pi \\ \circ \end{array} \right] \left[\begin{array}{c} \theta \\ \circ \end{array} \right] \left[\begin{array}{c} \bullet \\ \bullet \end{array} \right] = e^{i\theta}, \quad \left[\begin{array}{c} \alpha \\ \circ \end{array} \right] = 1 + e^{i\alpha}, \quad \left[\begin{array}{c} \bullet \\ \bullet \end{array} \right] \left[\begin{array}{c} \bullet \\ \bullet \end{array} \right] = \frac{1}{2}, \quad \left[\begin{array}{c} \circ \\ \circ \end{array} \right] = 2. \quad (2.11)$$

Furthermore,

$$\cos(\theta) = \frac{e^{i\theta} + e^{-i\theta}}{2} = \frac{1}{2} (e^{-i\theta} (1 + e^{2i\theta})). \quad (2.12)$$

²See Definition 3.2 for a clarification of what exactly universal means.

³Here we use the fact that composition between ZX diagrams that are scalars is regular multiplication. This is a subtle implication of 2.5.

Thus we can write $\cos(\theta)$ as a ZX diagram. By the Archimedean property, for all $r \in \mathbb{R}$, there exists $k \in \mathbb{N}$ such that $2^k > |r|$. So we can write

$$r = 2^k \cos(\theta), \quad \theta = \arccos\left(\frac{r}{2^k}\right). \quad (2.13)$$

Then any real number can be written as a ZX diagram. Now, for any $c \in \mathbb{C}$, use the polar coordinates $c = \rho \cdot e^{i\theta}$ to obtain a ZX representation of c . Note that for $c = 0$, $\llbracket \pi \rrbracket = 1 - 1 = 0$. $\square$

THEOREM 2.15 The ZX-Calculus is exactly universal for quantum computation.

Proof. This follows by Theorem 2.14 and Lemma 2.13. $\square$

Furthermore, it is a well-known result [JPV20] that the rules in Table 1 are complete for quantum computation. That is, any identity that can be shown through the matrix representation can also be shown through ZX-Calculus. Equivalently, if two diagrams D, D' represent the same quantum computation, the diagrams can be transformed to one another using only the rules in Table 1. In fact, we may largely ignore scalar factors during this rewrite process. This is formalized below:

LEMMA 2.16 We may ignore scalar factors during ZX rewrites, provided that the desired scalar is restored at the end. This gives us the computation up to a global phase.

Proof. Given a quantum computation as a unitary U , we can obtain a diagram D where $\llbracket D \rrbracket = cU$, where $c \in \mathbb{C}$ is some non-zero scalar factor. We then have $\llbracket D \rrbracket \llbracket D \rrbracket^\dagger = c\bar{c}I$. Then

$$\frac{1}{|c|} \llbracket D \rrbracket = \frac{c}{|c|} U, \quad (2.14)$$

so we can recover U up to a global phase. $\square$

Computationally, it is often convenient to keep track of the product of all scalar factors in a diagram as a *number* rather than as a ZX diagram. We can use this extraneous data structure precisely because of Theorem 2.14, and as such we will recover the computation's global phase as well (unlike Lemma 2.16).

2.2 Clifford Circuits

DEFINITION 2.17 A **Clifford circuit** is a quantum circuit made of **Clifford gates**:⁴

$$\text{CNOT} := \begin{array}{c} \text{---} \circ \text{---} \\ | \\ \text{---} \bullet \text{---} \end{array}, \quad H := \text{---} \square \text{---} = \begin{array}{c} \text{---} \circ \text{---} \\ | \\ \text{---} \bullet \text{---} \end{array} \begin{array}{c} \text{---} \circ \text{---} \\ | \\ \text{---} \bullet \text{---} \end{array} \begin{array}{c} \text{---} \circ \text{---} \\ | \\ \text{---} \bullet \text{---} \end{array}, \quad S := \text{---} \circ \text{---}. \quad (2.15)$$

Furthermore, let H and S be known as the generators of the **local Clifford gates**.

⁴This is not the typical formal definition of Clifford circuits. However, we will soon show that this definition is equivalent to the usual one, i.e. as the circuits that normalize the Pauli group.

DEFINITION 2.18 A **Clifford ZX diagram** is a ZX diagram made up only of spiders whose phases are multiples of $\frac{\pi}{2}$. Furthermore, say a ZX diagram D is **unitary** if $\llbracket D \rrbracket = cU$ for some $c \in \mathbb{C}$ and unitary matrix U .

THEOREM 2.19 Clifford circuits are precisely the unitary Clifford ZX diagrams. That is, every Clifford circuit can be written as a unitary Clifford ZX diagram and vice versa.

Proof, part 1. It is clear that any Clifford circuit is a Clifford ZX diagram, since the diagrams of the Clifford gates only involve spiders with phases multiples of $\frac{\pi}{2}$. The reverse direction, however, will take more work. $\square$

To show the reverse direction, we construct an algorithm from Chapter 5 of [KW24] that allows us to convert any unitary Clifford ZX diagram into a Clifford circuit.

DEFINITION 2.20 A **Hadamard wire** is a wire with a single Hadamard, i.e. $\text{---}\square\text{---}$. To further disambiguate this from $\text{---}\bigcirc\text{---}$, we may write $\text{---}\text{---}$ in place of $\text{---}\square\text{---}$.

DEFINITION 2.21 A ZX diagram is **graph-like** when:

- (a) Every spider is a Z-spider.
- (b) Spiders are only connected via Hadamard edges.
- (c) There are no self loops or parallel i.e. duplicate edges.
- (d) Every Z-spider is connected to at most one input and at most one output via a regular edge.
- (e) Every input and output is connected to a Z-spider.

THEOREM 2.22 Every ZX diagram, up to scalar factors, can be efficiently rewritten as a graph-like diagram.

Proof. Let D be an arbitrary ZX diagram. Run the following algorithm:

1. Convert all X-spiders to Z-spiders via (cc) .
2. Cancel adjacent H with (hh) .
3. Fuse all connected spiders via (f) until this cannot be done.
4. Remove parallel Hadamard edges via:

$$\begin{array}{c} \text{---}\bigcirc\text{---}\square\text{---}\bigcirc\text{---} \\ \text{---}\alpha\text{---}\beta\text{---} \end{array} \stackrel{(cc)}{=} \begin{array}{c} \text{---}\bigcirc\text{---}\square\text{---}\bigcirc\text{---}\square\text{---}\bigcirc\text{---} \\ \text{---}\alpha\text{---}\beta\text{---} \end{array} \stackrel{(f)}{=} \begin{array}{c} \text{---}\bigcirc\text{---}\square\text{---}\bigcirc\text{---}\square\text{---}\bigcirc\text{---} \\ \text{---}\alpha\text{---}\beta\text{---} \end{array} \stackrel{(hf)}{=} \begin{array}{c} \text{---}\bigcirc\text{---}\square\text{---}\bigcirc\text{---}\square\text{---}\bigcirc\text{---} \\ \text{---}\alpha\text{---}\beta\text{---} \end{array} \stackrel{(f)}{=} \begin{array}{c} \text{---}\bigcirc\text{---}\square\text{---}\bigcirc\text{---}\square\text{---}\bigcirc\text{---} \\ \text{---}\alpha\text{---}\beta\text{---} \end{array} \stackrel{(cc)}{=} \begin{array}{c} \text{---}\bigcirc\text{---}\square\text{---}\bigcirc\text{---}\square\text{---}\bigcirc\text{---} \\ \text{---}\alpha\text{---}\beta\text{---} \end{array}. \quad (2.16)$$

5. Remove self-loops via:

$$\begin{array}{c} \text{Spider with self-loop } \alpha \\ \xrightarrow{(id)} \text{Spider with self-loop } \alpha \\ \xrightarrow{(f)} \text{Spider with } \alpha \end{array}, \quad (2.17)$$

$$\begin{array}{c} \text{Spider with self-loop } \alpha \\ \propto \text{Spider with self-loop } \alpha + \pi \\ \xrightarrow{(hf)} \text{Spider with } \alpha + \pi \end{array}. \quad (2.18)$$

6. Introduce Z-spiders and Hadamards using (id) and (hh) to ensure every input and output is connected to a Z-spider and no Z-spiders are connected to multiple inputs or multiple outputs.

After this, all spiders are Z-spiders. All edges between spiders are Hadamard edges, else the spiders would have been fused together. There is at most one edge between any two spiders, and there are no self-loops. Step 6 ensures (e) is satisfied. $\square$

Note that the above procedure applies to any ZX diagram, not just Clifford ones. As it turns out, when our starting ZX diagram is Clifford, we can simplify the diagram even further. To see how, we introduce the notion of a graph state and some operations on them.

DEFINITION 2.23 Say a spider in a ZX diagram is **internal** if it is not directly connected to an input or output wire, but it has a connection via other spiders to an input or output wire. Conversely, a spider is a **boundary** spider if it is directly connected to an input or output wire.

DEFINITION 2.24 For a simple, undirected graph $G = (V, E)$, the **graph state** $|G\rangle$ of G is defined

$$|G\rangle := \prod_{(v,w) \in E} CZ_{v,w} |+\rangle^{\otimes |V|}. \quad (2.19)$$

That is, prepare the $|+\rangle^{\otimes |V|}$ state and apply a CZ gate between every pair of adjacent vertices in G . If D is a ZX diagram and graph state, we write $G(D) = (V, E)$ to refer to the graph. Note that in the ZX Calculus,

$$|+\rangle \propto \text{circle with a dot}, \quad CZ = \text{two circles connected by a vertical line}. \quad (2.20)$$

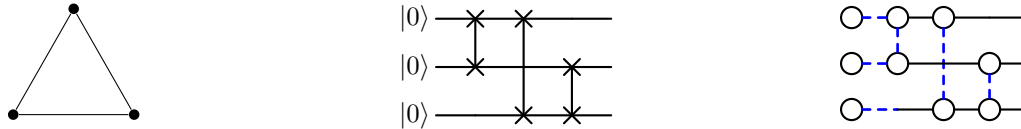


Figure 1: The triangle graph and its circuit and ZX graph state.

DEFINITION 2.25 A **graph state with local Cliffords (GSLC)** is a graph state to which some single-qubit Clifford unitaries have been applied on its outputs.

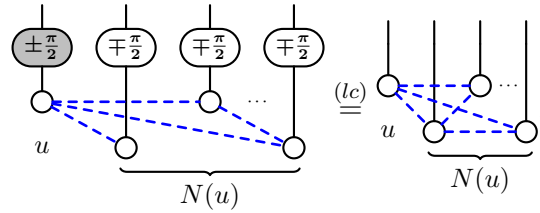
THEOREM 2.26 Every Clifford ZX diagram can be written in GSLC form.

To prove Theorem 2.26, we need to show that all internal spiders in a Clifford ZX diagram in graph-like form can be removed, perhaps by changing the phases of some of the boundary spiders and adding local Cliffords. To do this we need the following graph operations.

DEFINITION 2.27 Let G be a graph and $u \in V(G)$. The **local complementation** of G around u , written $G \star u$, is the graph obtained by complementing the subgraph induced by the neighbors of u . For distinct $v, w \in V(G)$, let $e_{v,w}$ be the edge between them. Then

$$e_{v,w} \in E(G \star u) \iff \begin{cases} e_{v,w} \notin E(G) & \text{if } v, w \in N_G(u), \\ e_{v,w} \in E(G) & \text{otherwise.} \end{cases} \quad (2.21)$$

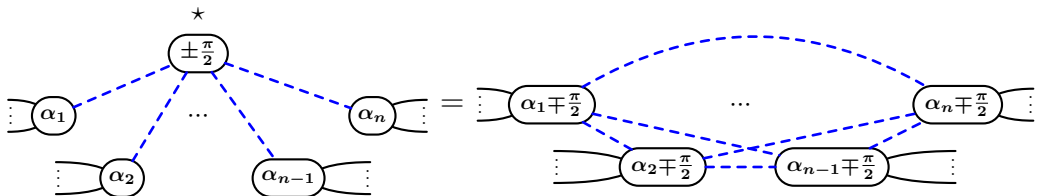
LEMMA 2.28 Let D be a graph-state ZX diagram, and u some vertex in $G(D)$. Then we can obtain a diagram D' with $G(D') = G(D) \star u$ by applying a $R_X(-\pi/2)$ gate on the spider corresponding to u and $R_Z(\pi/2)$ on all spiders in $N_{G(D)}(u)$. Call this rule (lc) :


(2.22)

Thus by applying local Cliffords we can local complement a graph state.

Proof. This was originally proved in Theorem 2 of [DP09]. The proof surmounts to a clever decomposition of the H-box as Z and X-spiders, which after some application of the rules in Table 1 gives the desired result. $\square$

LEMMA 2.29 The following holds:


(2.23)

Here, the $\star$ above $(\pm\frac{\pi}{2})$ indicates a local complementation at this vertex.

Proof. We provide a proof sketch: Let u denote the vertex we local complement around. First, push out all the phases via (f) to obtain a graph state where all the vertices are phase-free Z-spiders. Apply local complementation in reverse around u . Then u will have $(\pm\frac{\pi}{2}) - (\pm\frac{\pi}{2}) = \bullet$ dangling from it. Apply (cp) to push the $\bullet$ through and disconnect u . This gives the RHS. For a step-by-step proof, see Appendix B.3 of [Dun+20]. $\square$

DEFINITION 2.30 Let G be a graph, $u, v \in V(G)$, and $e_{uv} \in E(G)$. The pivot of G along uv , denoted $G \wedge uv$, is defined as

$$G \wedge uv := G \star u \star v \star u \quad (2.24)$$

or equivalently

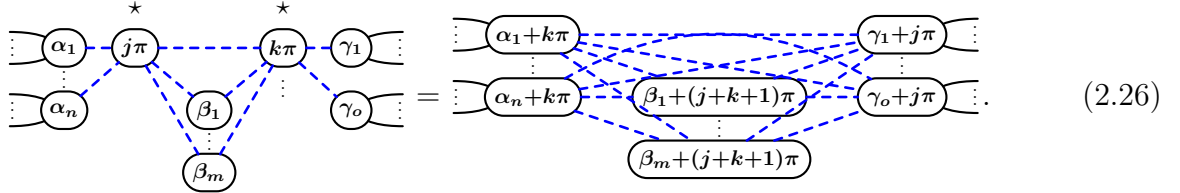
$$G \wedge uv = G \star v \star u \star v. \quad (2.25)$$

Proof. This is a standard proof by cases, seen for example in Definition 4.3 of [Dun+20]. $\square$

LEMMA 2.31 Given a graph G , we can pivot around uv for $u, v \in V(G)$ by applying Hadamard on u and v and Z on all vertices in $N_G(u) \cap N_G(v)$, inducing a swap gate between u and v .

Proof. The proof is surprisingly involved and can be found in Proposition 3.1 of [DP14]. $\square$

LEMMA 2.32 The following holds:

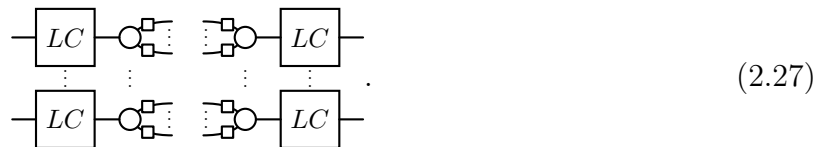


Note that the vertices $\beta_1, \dots, \beta_m$ may have other wires, but we suppress them here to reduce clutter.

Proof. We provide a proof sketch: Let u denote the vertex with phase $j\pi$ and v the vertex with phase $k\pi$. First, push out all the phases via (f) to obtain a graph state where all the vertices are phase-free Z -spiders. Apply pivoting on uv . Then u will have $(k\pi) \text{---} \square \text{---}$ dangling from it (note the change from j to k due to the pivoting's swap gate), and similarly for v . Apply (cc) and (cp) to obtain the RHS after fusing with (f) . For a step-by-step proof, see Appendix B.3 of [Dun+20]. $\square$

Proof of Theorem 2.26. First note that by Theorem 2.11, we can topologically morph the inputs of the Clifford circuit to face rightwards, like outputs, and that this has no effect on the operation represented by the diagram. Now, start by converting a Clifford ZX diagram to graph-like form. Apply Lemma 2.29 and 2.32 to remove all internal spiders from the diagram, perhaps appending local Clifford gates to the output wires. With some care it can be shown that this process terminates. This gives a circuit in GSLC form. $\square$

LEMMA 2.33 The GSLC form of a CLifford ZX diagram may be written as follows:



Here $\boxed{LC}$ refers to circuits of the form $H^i S^j$ for some $i \in \{0, 1\}$ and $j \in \{0, 1, 2, 3\}$.

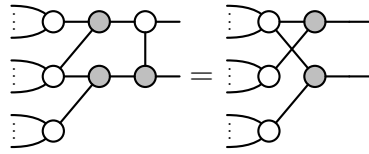
Proof. First note that we may topologically morph the GSLC form such that input wires face left and output wires face right. Now, note that the local Clifford generated must be of the form $H^i S^j$. Namely, when applying Lemma 2.29, we only modify the phase of boundary spiders. When we apply Lemma 2.32, we may add a Hadamard gate to an input or output wire. Since two Hadamards cancel, each input or output wire has at most one Hadamard. Since the phase of each boundary spider is a multiple of $\frac{\pi}{2}$ modulo 2π , unfusing the phase requires at most three S gates. $\square$

LEMMA 2.34 Let U be a Clifford unitary. Then U can be written as a circuit consisting of 8 layers respectively made up of only the following gates:

$$H \quad S \quad CZ \quad CNOT \quad H \quad CZ \quad S \quad H. \quad (2.28)$$

Proof. This proof is taken from Theorem 5.3.11 of [KW24]. We provide a proof sketch: Start from Lemma 2.33. This explains the outer H and S layers. Now, apply (f) in reverse to obtain a CZ, representing an edge between two vertices both on the left or both on the right. This explains the two CZ layers. Then, apply (cc) to obtain the H layer in between CNOT and CZ.

What remains is a bipartite graph whose left vertices are phase-less Z-spiders and right vertices phase-less X-spiders. As it turns out, because our diagram is unitary, this bipartite graph can be decomposed into a circuit made only of CNOT gates via the following identity:

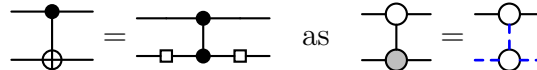


$$(2.29)$$

The proof of this surmounts to book-keeping and a clever use of (b). Please see Lemma 4.2.11 of [KW24] for a step-by-step proof. $\square$

Proof of Theorem 2.26 and Proof of Theorem 2.19, part 2. Given a unitary Clifford ZX diagram, do the following:

1. Convert it to graph-like form, using Theorem 2.22.
2. Use Lemma 2.29 and Lemma 2.32 to remove internal spiders and bring the diagram to GSLC form.
3. Use Lemma 2.34 to obtain a Clifford circuit for the diagram, by noting that



$$(2.30)$$

Thus the proof of Theorem 2.19 is complete. $\square$

COROLLARY 2.35 Any n -qubit Clifford unitary can be represented by an equivalent circuit in at most $4n(n + 3/2) - 1$ Hadamard, S, and CNOT gates.

Proof. Look at Lemma 2.34. Each Hadamard layer has at most n gates, and there are three of those. Each S gate layer has at most $3n$ gates, and there are two of them. A circuit of CZ gates can have at most one CZ gate for every pair of qubits, as CZ gates commute past each other. Thus the CZ layer has at most $n(n-1)/2$ gates. CZs are made up of one CNOT and two Hadamards, so this contributes $3 \cdot n(n-1)/2$ gates. Finally, the CNOT circuit is essentially the number of distinct invertible linear transformations over $\mathbb{F}_2^n$. By Gaussian elimination, any such operation can be synthesized using $n^2 - 1$ CNOT gates. Thus, in total at most

$$9n + 2 \cdot 3 \cdot \frac{n(n-1)}{2} + n^2 - 1 = 4n(n + 3/2) - 1 \quad (2.31)$$

gates are used. $\square$

THEOREM 2.36 Let U be a q -qubit Clifford unitary with N gates. We can rewrite this as a ZX diagram in GSLC form in $O(Nq^2)$ time, if $N \geq q$. Further, we can calculate any amplitude

$$\langle x_1, \dots, x_q | U | 0, \dots, 0 \rangle \quad (2.32)$$

in $O(Nq^2)$ time. This is practically equivalent to the **Gottesman-Knill theorem**. If U is already available in GSLC form, this can be done in $O(q^3)$ time.

Proof. The first comes from analyzing the runtime of the algorithm in Theorem 2.19's part 2 proof. The second comes from the fact that, a post-selection of outcomes $|x_1, \dots, x_n\rangle$ is equivalent to “plugging” the output wires of a GSLC diagram with the corresponding spiders. The third claim follows from a similar observation. Please see Proposition 5.4.4 of [KW24] for a more detailed proof. $\square$

2.3 The Stabilizer Formalism

When we view the Clifford circuits of the previous section more algebraically, we find interesting connections to the following objects:

DEFINITION 2.37 The **Pauli matrices** or **Pauli gates** refer to the following gates:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \quad (2.33)$$

DEFINITION 2.38 Say a matrix A is a **stabilizer** of the state $|\psi\rangle$ when $|\psi\rangle$ is a $+1$ eigenvector of A , i.e. $A|\psi\rangle = |\psi\rangle$.

DEFINITION 2.39 The n -qubit Pauli group is defined as follows:

$$\mathcal{P}_n := \{i^k P_1 \otimes \dots \otimes P_n \mid k \in \{0, 1, 2, 3\}, P_j \in \{I, X, Y, Z\}\}. \quad (2.34)$$

Call the elements of $\mathcal{P}_n$ as **Pauli strings**, written as vectors of length n :

$$\vec{P} := i^k P_1 \otimes \dots \otimes P_n. \quad (2.35)$$

Proof. We need to show that this is indeed a group. Associativity follows from matrix multiplication. I is the identity. Each Pauli matrix is its own inverse, i.e. $X^2 = Y^2 = Z^2 = 1$, and it is easy to see that the inverses of $i^k P$ exist for all k and P . Finally, since two non-identity Pauli matrices $P_1 \neq P_2$ anti-commute, i.e. $P_1 P_2 = -P_2 P_1$, an exhaustive multiplication table shows the group is closed and finite. $\square$

THEOREM 2.40 For any Clifford unitary U and Pauli string $\vec{P}$ of compatible dimensions, there exists a pauli string Q such that $U\vec{P}U^\dagger = \vec{Q}$.

Proof. It suffices to check that, when we commute a basis of the Pauli strings through the Hadamard, S, and CNOT gate, we will end up with another Pauli string. Notice that $\{X, Z\}$ is a basis for $\mathcal{P}_1$, up to scalar factors. Now note that

$$\text{---} \circ \pi \text{---} \square \stackrel{(cc)}{=} \text{---} \square \text{---} \circ \pi \text{---}, \quad \text{---} \circ \pi \text{---} \square \stackrel{(cc)}{=} \text{---} \square \text{---} \circ \pi \text{---}. \quad (2.36)$$

For the S gate,

$$\text{---} \circ \pi \text{---} \circ \frac{\pi}{2} \text{---} \stackrel{(f)}{=} \text{---} \circ \frac{\pi}{2} \text{---} \circ \pi \text{---}, \quad \text{---} \circ \pi \text{---} \circ \frac{\pi}{2} \text{---} \stackrel{\begin{smallmatrix} (\pi) \\ (f) \end{smallmatrix}}{=} \text{---} \circ \frac{\pi}{2} \text{---} \circ \pi \text{---} \circ \pi \text{---}. \quad (2.37)$$

For the CNOT gate,

$$\begin{array}{ccc} \text{---} \circ \pi \text{---} \circ \text{---} & \stackrel{(f)}{=} & \text{---} \circ \text{---} \circ \pi \text{---} \\ \text{---} \circ \text{---} & & \text{---} \circ \text{---} \end{array}, \quad \begin{array}{ccc} \text{---} \circ \pi \text{---} \circ \text{---} & \stackrel{\begin{smallmatrix} (\pi) \\ (f) \end{smallmatrix}}{=} & \text{---} \circ \text{---} \circ \pi \text{---} \\ \text{---} \circ \text{---} & & \text{---} \circ \pi \text{---} \end{array}, \quad (2.38)$$

$$\begin{array}{ccc} \text{---} \circ \text{---} \circ \text{---} & \stackrel{\begin{smallmatrix} (\pi) \\ (f) \end{smallmatrix}}{=} & \text{---} \circ \text{---} \circ \pi \text{---} \\ \text{---} \circ \pi \text{---} & & \text{---} \circ \pi \text{---} \end{array}, \quad \begin{array}{ccc} \text{---} \circ \text{---} \circ \text{---} & \stackrel{(f)}{=} & \text{---} \circ \text{---} \circ \text{---} \\ \text{---} \circ \pi \text{---} & & \text{---} \circ \pi \text{---} \end{array}. \quad \square$$

DEFINITION 2.41 For a set $\mathcal{S} \subseteq \mathcal{P}_n$, define the **stabilizer subspace**⁵ of $\mathcal{S}$ as

$$\text{Stab}(\mathcal{S}) := \{|\psi\rangle \mid \vec{P}|\psi\rangle = |\psi\rangle, \forall \vec{P} \in \mathcal{S}\}. \quad (2.39)$$

If $\text{Stab}(\mathcal{S}) \neq \{0\}$, call $\mathcal{S}$ a **stabilizer group**.

Proof. If A, B are stabilizers of $|\psi\rangle$, then $AB|\psi\rangle = A|\psi\rangle = |\psi\rangle$ so AB is also a stabilizer. Furthermore,

$$A|\psi\rangle = |\psi\rangle \implies |\psi\rangle = A^{-1}|\psi\rangle \quad (2.40)$$

so A^{-1} is also a stabilizer of $|\psi\rangle$. Thus $\mathcal{S}$ is a subgroup. On the other hand, if two states $|\psi\rangle$ and $|\phi\rangle$ are stabilized by A , then

$$A(\lambda|\psi\rangle + \mu|\phi\rangle) = \lambda A|\psi\rangle + \mu A|\phi\rangle = \lambda|\psi\rangle + \mu|\phi\rangle, \quad (2.41)$$

for any $\lambda, \mu \in \mathbb{C}$. Since 0 is trivially in every $\text{Stab}(\mathcal{S})$, this is indeed a subspace. $\square$

⁵This is a strictly more restrictive definition of stabilizer subspaces and stabilizer groups than the definitions found in algebra, for instance. However, since we are only dealing with pure states, and are only concerned with the stabilizer subspaces of $\mathcal{P}_n$, this definition suffices.

LEMMA 2.42 If a subgroup $\mathcal{S} \subseteq \mathcal{P}_n$ is a stabilizer subgroup, then:

1. $-I \notin \mathcal{S}$,
2. $\vec{P}^2 = I$ for all $\vec{P} \in \mathcal{S}$, and
3. $\mathcal{S}$ is commutative.

Proof. If $-I \in \mathcal{S}$, then all $|\psi\rangle \in \text{Stab}(\mathcal{S})$ satisfies $|\psi\rangle = -I|\psi\rangle = -|\psi\rangle$ such that $\text{Stab}(\mathcal{S}) = \{0\}$, a contradiction. An exhaustive multiplication table for $\mathcal{P}_1$ reveals that if $\vec{P}^2 \neq I$, then $\vec{P}^2 = -I$ so again $\text{Stab}(\mathcal{S}) = \{0\}$. Thus, each $P \in \mathcal{S}$ must have $P^2 = I$. Then $P^\dagger = P$, since $P^{-1} = P$ and its phase cannot be $\pm i$. Then for any $P, Q \in \mathcal{S}$, $PQ = (PQ)^\dagger = Q^\dagger P^\dagger = QP$. $\square$

THEOREM 2.43 This is Theorem 1 of [AG04]. Given an n -qubit state $|\psi\rangle$, the following are equivalent:

1. $|\psi\rangle$ can be obtained from $|0\rangle^{\otimes n}$ by CNOT, H, and S gates only.
2. $|\psi\rangle$ can be obtained from $|0\rangle^{\otimes n}$ by CNOT, H, S, and measurement gates only.
3. $|\psi\rangle$ is stabilized by exactly 2^n Pauli operators.
4. $|\psi\rangle$ is uniquely determined by $\text{Stab}(|\psi\rangle)$.

Thus, we can call any circuit consisting entirely of Clifford gates and measurements a **stabilizer circuit**, and any state obtainable by applying a stabilizer circuit to $|0\rangle^{\otimes n}$ a **stabilizer state**.

Proof. A concise proof is available in [AG04], which itself relies on the Gottesman-Knill theorem (Theorem 2.36), whose ZX-proof can be found in Chapter 5 of [KW24] and non-ZX-proof in the original paper [Got98]. $\square$

LEMMA 2.44 Note that $|\mathcal{P}_n| = 4^{n+1}$. Furthermore, the number of stabilizer states on n qubits is:

$$2^n \prod_{i=0}^{n-1} (2^{n-i} + 1). \quad (2.42)$$

Proof. The first claim follows almost immediately from Definition 2.39. By Theorem 2.43, to obtain the second claim we can equivalently count the number of ordered generating sets for stabilizer groups, divided by the number of generating sets for a given stabilizer. This is the proof used in Proposition 2 of [AG04]. $\square$

Informally, the stabilizer formalism tells us that the states obtainable by applying Clifford circuits to $|0\rangle^{\otimes n}$ are exactly the pure stabilizer states. This is useful because stabilizer states admit a compact classical description. The tableau formalism below is one such description: it keeps track of how a Clifford circuit conjugates a convenient set of Pauli generators.

DEFINITION 2.45 Let $|\psi\rangle$ be an n -qubit stabilizer state with stabilizer group $\mathcal{S}(|\psi\rangle)$. A **tableau** for $|\psi\rangle$ is a binary matrix encoding $2n$ signed Pauli strings

$$R_i = (-1)^{r_i} P_{i,1} \otimes \cdots \otimes P_{i,n}, i \in [2n]. \quad (2.43)$$

A tableau has the form

$$\left[\begin{array}{ccc|ccc|c} x_{11} & \cdots & x_{1n} & z_{11} & \cdots & z_{1n} & r_1 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{n1} & \cdots & x_{nn} & z_{n1} & \cdots & z_{nn} & r_n \\ \hline x_{(n+1)1} & \cdots & x_{(n+1)n} & z_{(n+1)1} & \cdots & z_{(n+1)n} & r_{n+1} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{(2n)1} & \cdots & x_{(2n)n} & z_{(2n)1} & \cdots & z_{(2n)n} & r_{2n} \end{array} \right], \quad (2.44)$$

where

$$x_{ij} = \begin{cases} 1 & \text{if } P_{i,j} = X \text{ or } P_{i,j} = Y, \\ 0 & \text{otherwise,} \end{cases} \quad z_{ij} = \begin{cases} 1 & \text{if } P_{i,j} = Z \text{ or } P_{i,j} = Y, \\ 0 & \text{otherwise.} \end{cases} \quad (2.45)$$

The bit r_i records the sign of the Pauli string:

$$r_i = \begin{cases} 0 & \text{if } R_i \text{ has sign } +1, \\ 1 & \text{if } R_i \text{ has sign } -1. \end{cases} \quad (2.46)$$

The bottom n rows are stabilizer generators for $|\psi\rangle$. The top n rows are **destabilizer generators**. They do not stabilize $|\psi\rangle$. Rather, they are auxiliary Pauli strings chosen so that, together with the stabilizer generators, they form a convenient *symplectic* basis of the Pauli group. In the Aaronson–Gottesman tableau, the destabilizers are what make the row operations and measurements efficient and canonical.

THEOREM 2.46 Using the tableau representation, Clifford gates such as H , S , and CNOT can be simulated efficiently by updating the binary entries of the tableau. Pauli measurements can also be simulated efficiently using the same representation. This gives the tableau-based form of the Gottesman–Knill simulation algorithm [AG04].

Proof intuition. The key observation is that each $P_{i,j}$ may be encoded by two bits:

$$\begin{array}{c|cc} P & x & z \\ \hline I & 0 & 0 \\ X & 1 & 0 \\ Z & 0 & 1 \\ Y & 1 & 1 \end{array} \quad (2.47)$$

Thus an n -qubit Pauli string can be encoded by $2n$ bits, together with one additional sign bit. This is possible since stabilizer generators never have phases $\pm i$ by Lemma 2.42.

A Clifford gate maps Pauli strings to Pauli strings under conjugation. Hence, to update the state after applying a Clifford gate, it suffices to update how that gate conjugates each

row of the tableau. For example, applying H to qubit j swaps the x_{ij} and z_{ij} entries in column j , up to the appropriate sign change when Y is involved. Similarly, S and CNOT act by simple binary updates on the entries. Thus the tableau is best viewed not as a list of stabilizers, but as an algorithmic data structure for simulating stabilizer computations. $\square$

Note that this algorithm may be improved further. See [SSY25] for such an example.

3 Measurement-Based Quantum Computation

3.1 Universality

In this thesis, we are primarily concerned with the universality of the comparability grid for measurement-based quantum computation. It follows that we need a precise definition of universality to formally state our problem.

DEFINITION 3.1 For every $n \in \mathbb{N}$, let $SU(n)$ denote the **special unitary group** under matrix multiplication consisting of $n \times n$ unitary matrices with determinant 1. It is a normal subgroup of the **unitary group** $U(n)$, the group of $n \times n$ unitary matrices.

DEFINITION 3.2 A set of gates $\mathcal{S}$ is (efficiently) **universal** if, for every $n \in \mathbb{N}$, every $U \in SU(2^n)$ can be generated (in $\text{poly}(n)$ gates) by gates in $\mathcal{S}$ [Nes+07]. We say a set of gates $\mathcal{S}$ is **d -qubit universal** if the above holds for $U \in SU(2^d)$.

In this thesis we will only focus on universality, not efficient universality. Thus, simulating certain unitaries may require super-polynomially many gates. Also, in practice, the application of U up to some metric and error ϵ suffices. This is called **approximate universality**. This thesis will focus solely on **exact universality**, where we desire no error, rather than approximate universality. Henceforth, all references to universality will refer to exact, and not necessarily efficient, universality.

Furthermore, note that Definition 3.2 is only concerned with implementing arbitrary unitaries, not necessarily arbitrary computation. That is, it does not specify whether the inputs and outputs are classical or quantum in nature. This matters because $|0\rangle$ and $-|0\rangle$ have the same measurement distribution and classical output, but $|0\rangle \neq -|0\rangle$. This difference is called **global phase**, i.e. a scalar $e^{i\theta}$ multiplied to an entire state.

DEFINITION 3.3 We use the following types of **universal computation** from [Nes+07]:

- **CC-universal:** For every $U \in SU(2^n)$, the model can reproduce the distribution of local measurements performed on $U|0^n\rangle$.
- **QC-universal:** Given an arbitrary n -qubit state $|\phi\rangle$, for every $U \in SU(2^n)$, the model can reproduce the distribution of local measurements performed on $U|\phi\rangle$.
- **CQ-universal:** For every $U \in SU(2^n)$, the model can produce the quantum state $|\phi\rangle = U|0^n\rangle$.
- **QQ-universal:** Given an arbitrary n -qubit state $|\phi\rangle$, for every $U \in SU(2^n)$, the model can produce the quantum state $U|\phi\rangle$.

For this thesis, we will focus on CQ-universality, out of convenience and convention.⁶ This distinction turns out to be quite important for algorithmic and implementation reasons, as we will see in Section 5. Henceforth, universality shall refer to (exact, not necessarily efficient) CQ-universality, unless otherwise noted.

⁶Many papers which examine the universality of MBQC resource states really only examine CQ-universality. For example, [Nes+07].

3.2 The One-Way Model

DEFINITION 3.4 Informally, **measurement-based quantum computation** (MBQC) is a deterministic⁷ protocol for quantum computation that roughly goes as follows:

1. A classical description specifies the desired computation.
2. A resource state $|\phi_k\rangle$ is chosen from some family of resource states $\Phi = \{|\phi_1\rangle, |\phi_2\rangle, \dots\}$. A graph state is an example of a resource state.
3. Measurements are done on the qubits of $|\phi_k\rangle$ in an adaptive fashion. Adaptive means that a classical computer may decide to change the next measurement's basis depending on a previous measurement outcome.
4. By the end, the remaining unmeasured qubits contain the desired output state, up to known Pauli errors.

We will make this definition more precise after we develop some more notation.

EXAMPLE 3.5 We start by considering the simplest possible MBQC circuit. Define the **Hadamard gadget** as the following circuit:

$$\begin{array}{c} |\phi\rangle \\ |+\rangle \end{array} \begin{array}{c} \bullet \\ \bullet \end{array} \text{---} \boxed{H} \text{---} \boxed{\text{meter}} . \quad (3.1)$$

For now, we will focus on what this circuit does. Commuting a Hadamard through the CZ gives a CNOT:

[illegible]

Two CNOTs cancel each other out and a CNOT's target acts trivially on $|+\rangle$. So

$$\begin{array}{c} H|\phi\rangle \\ |+\rangle \end{array} \begin{array}{c} \text{---} \oplus \text{---} \\ | \\ \text{---} \bullet \text{---} \end{array} \boxed{\text{meter}} = \begin{array}{c} H|\phi\rangle \\ |+\rangle \end{array} \begin{array}{c} \text{---} \bullet \text{---} \oplus \text{---} \bullet \text{---} \bullet \text{---} \\ | \\ \text{---} \oplus \text{---} \bullet \text{---} \oplus \text{---} \oplus \text{---} \end{array} \boxed{\text{meter}}. \quad (3.3)$$

The first three CNOTs are equivalent to a swap gate. Thus

$$\begin{array}{c}
H|\phi\rangle \\
|+\rangle
\end{array}
\begin{array}{c}
\bullet \\
\oplus \\
\bullet \\
\oplus
\end{array}
\begin{array}{c}
\oplus \\
\bullet \\
\oplus \\
\oplus
\end{array}
\begin{array}{c}
\bullet \\
\bullet \\
\bullet \\
\bullet
\end{array}
\begin{array}{c}
\boxed{\text{meter}} \\
\text{---} \\
\text{---}
\end{array}
=
\begin{array}{c}
|+\rangle \\
H|\phi\rangle
\end{array}
\begin{array}{c}
\bullet \\
\oplus
\end{array}
\begin{array}{c}
\boxed{\text{meter}} \\
\text{---}
\end{array}. \quad (3.4)$$

⁷Here, deterministic means that all measurement branches implement the same intended transformation after the appropriate correction

The final state before measurement is

$$\text{CNOT}(|+\rangle \otimes H|\phi\rangle) = \frac{1}{\sqrt{2}}(|0\rangle \otimes H|\phi\rangle + |1\rangle \otimes XH|\phi\rangle). \quad (3.5)$$

In other words, a measurement of the first qubit in the **computational basis** ($\{|0\rangle, |1\rangle\}$) will tell us whether we have a $H|\phi\rangle$ or $XH|\phi\rangle$ on the output wire.

Now, suppose our intention in the above circuit was to apply a Hadamard gate to $|\phi\rangle$. Then there is a $\frac{1}{2}$ probability that we apply XH instead of H , which is indicated when we measure a 1. How can we get rid of this additional X ?

EXAMPLE 3.6 Define the **teleportation gadget** as the following circuit.

$$\begin{array}{c} |\phi\rangle \\ |+\rangle \end{array} \begin{array}{c} \bullet \\ \bullet \end{array} \begin{array}{c} \text{---} R_Z(\theta) \text{---} H \text{---} \text{Measurement} \end{array} = \begin{array}{c} |+\rangle \\ HR_Z(\theta)|\phi\rangle \end{array} \begin{array}{c} \bullet \\ \oplus \end{array} \text{Measurement}. \quad (3.6)$$

Let $m \in \{0, 1\}$ be the measurement outcome of measuring in the computational basis. Then we apply $X^m H R_Z(\theta)$ to $|\phi\rangle$.

Proof. Note that, by (f), the $R_Z(\theta)$ gate commutes through the CZ unchanged. The rest of the analysis is identical to Example 3.5. $\square$

EXAMPLE 3.7 Now, suppose we chain together two teleportation gadgets, taking the output of one as the input $|\phi\rangle$ to another. Then we are applying the following gate:

$$X^{m_2} H R_Z(\theta_2) X^{m_1} H R_Z(\theta_1) \stackrel{(\pi)}{=} X^{m_2} H X^{m_1} R_Z((-1)^{m_1} \theta_2) H R_Z(\theta_1) \quad (3.7)$$

$$\stackrel{(cc)}{=} X^{m_2} Z^{m_1} H R_Z((-1)^{m_1} \theta_2) H R_Z(\theta_1) \quad (3.8)$$

Chaining three gadgets, we are applying the following gate:

$$X^{m_3} H R_Z(\theta_3) X^{m_2} H R_Z(\theta_2) X^{m_1} H R_Z(\theta_1) \quad (3.9)$$

$$= X^{m_3} H R_Z(\theta_3) (X^{m_2} Z^{m_1} H R_Z((-1)^{m_1} \theta_2) X^{m_1} H R_Z(\theta_1)) \quad (3.10)$$

$$\stackrel{(\pi)}{\stackrel{(cc)}}{=} X^{m_3+m_1} Z^{m_2} (H R_Z((-1)^{m_2} \theta_3) H R_Z((-1)^{m_1} \theta_2) X^{m_1} H R_Z(\theta_1)). \quad (3.11)$$

Inductively, since $X^2 = Z^2 = I$ and Pauli operators remain Pauli operators under conjugation by H and $R_Z(\theta)$ up to the corresponding angle updates, chaining any number of gadgets produces an overall Pauli byproduct $X^m Z^n$ for some bits m, n .

The formal definition of the MBQC protocol incorporates these errors into the definition.

DEFINITION 3.8 We use the formal definition from [Bro+07], known as the **Measurement Calculus**. Denote an arbitrary single qubit state by

$$|+\theta, \phi\rangle := \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right)|1\rangle \quad (3.12)$$

and denote its orthogonal state by

$$|-\theta, \phi\rangle := \sin\left(\frac{\theta}{2}\right)|0\rangle - e^{i\phi}\cos\left(\frac{\theta}{2}\right)|1\rangle \quad (3.13)$$

where $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$. A computation in this model is a series of the following deterministic **commands**:

1. Preparation: 1-qubit preparations N_i , to prepare qubit i in the state $|+\rangle$;
2. Entanglement: 2-qubit entangling operators $E_{i,j} := CZ_{i,j}$;
3. Measurement: 1-qubit measurements $M_i^{\lambda, \alpha}$ on planes $\lambda \in \{(X, Y), (X, Z), (Y, Z)\}$, defined as projections into

- (i) $|\pm_{(X,Y), \alpha}\rangle := |\pm_{\pi/2, \alpha}\rangle$ for $\lambda = (X, Y)$,
- (ii) $|\pm_{(X,Z), \alpha}\rangle := |\pm_{\alpha, 0}\rangle$ for $\lambda = (X, Z)$, and
- (iii) $|\pm_{(Y,Z), \alpha}\rangle := |\pm_{\alpha, \pi/2}\rangle$ for $\lambda = (Y, Z)$,

with the convention that $|+\theta, \phi\rangle\langle +\theta, \phi|$ corresponds to the outcome 0 and $|-\theta, \phi\rangle\langle -\theta, \phi|$ corresponds to the outcome 1; and

4. Correction: 1-qubit Pauli corrections X_i and Z_i .

The measurements $M_i^{\lambda, \alpha}$ may depend on **signals**, namely sums of some outcomes of previous measurements mod 2. These represent the Pauli errors of the form $X^m Z^n$ being carried through, for some bits m, n . A computation can then be expressed as a series of preparation, entanglement, measurement, and correction commands, strictly in this order, satisfying the following conditions:

- (a) No command depends on an outcome not yet measured;
- (b) No command acts on a qubit already measured;
- (c) No command acts on a qubit not yet prepared, unless it is an input qubit; and
- (d) A qubit i is measured if and only if i is not an output.

Finally, a **measurement pattern** $\mathcal{B} = (V, I, O, S)$ consists of the following data, satisfying conditions (a)–(d):

- a finite set of qubits $V \subseteq V(G)$,
- an input set $I \subseteq V$,
- an output set $O \subseteq V$, and
- a finite sequence S of measurement and correction commands.

Note that the measurement bases available in the Measurement Calculus model are different from the measurements in Examples 3.5 and 3.6. Thus, it will be convenient to find the operations applied by measurements in the XY, YZ, and XZ planes.

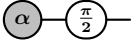
LEMMA 3.9 Let $|\psi\rangle = a|0\rangle + b|1\rangle$ be an arbitrary one-qubit input state. Prepare a second qubit $|+\rangle$, apply CZ, and project the first qubit onto the state

$$|m\rangle = u|0\rangle + v|1\rangle. \quad (3.14)$$

Conditioned on this measurement outcome, the unnormalized output state on the second qubit is

$$H \begin{bmatrix} \bar{u} & 0 \\ 0 & \bar{v} \end{bmatrix} |\psi\rangle. \quad (3.15)$$

In particular, for the standard measurement planes we obtain:

Plane	Measurement state	ZX effect	Unnormalized output	Unitary?
XY	$\frac{1}{\sqrt{2}}(0\rangle + e^{-i\alpha} 1\rangle)$		$\frac{1}{\sqrt{2}}HP(\alpha) \psi\rangle$	Always
YZ	$c 0\rangle + is 1\rangle$		$H \begin{bmatrix} c & 0 \\ 0 & -is \end{bmatrix} \psi\rangle$	Only if $ c = s $
XZ	$c 0\rangle + s 1\rangle$		$H \begin{bmatrix} c & 0 \\ 0 & s \end{bmatrix} \psi\rangle$	Only if $ c = s $

where $P(\alpha) = \text{diag}(1, e^{i\alpha}) = \llbracket \textcircled{\alpha} \rrbracket$, $c = \cos(\frac{\alpha}{2})$, and $s = \sin(\frac{\alpha}{2})$.

Proof. Write $|\psi\rangle = a|0\rangle + b|1\rangle$. Prepare a second qubit $|+\rangle$ and apply CZ:

$$CZ(|\psi\rangle|+\rangle) = CZ((a|0\rangle + b|1\rangle)|+\rangle) = a|0\rangle|+\rangle + b|1\rangle|-\rangle. \quad (3.16)$$

Now project the first qubit onto $|m\rangle = u|0\rangle + v|1\rangle$. The resulting unnormalized state is

$$(\langle m| \otimes I)CZ(|\psi\rangle|+\rangle) = (\langle m| \otimes I)(a|0\rangle|+\rangle + b|1\rangle|-\rangle) \quad (3.17)$$

$$= a\langle m|0\rangle|+\rangle + b\langle m|1\rangle|-\rangle \quad (3.18)$$

$$= a\bar{u}|+\rangle + b\bar{v}|-\rangle \quad (3.19)$$

$$= H \begin{bmatrix} \bar{u} & 0 \\ 0 & \bar{v} \end{bmatrix} |\psi\rangle. \quad (3.20)$$

For the XY-plane state, $|m\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{-i\alpha}|1\rangle)$,. So $\bar{u} = \frac{1}{\sqrt{2}}$, and $\bar{v} = \frac{e^{i\alpha}}{\sqrt{2}}$.. Hence

$$H \begin{bmatrix} \bar{u} & 0 \\ 0 & \bar{v} \end{bmatrix} |\psi\rangle = H \begin{bmatrix} 1/\sqrt{2} & 0 \\ 0 & e^{i\alpha}/\sqrt{2} \end{bmatrix} |\psi\rangle = \frac{1}{\sqrt{2}}H \begin{bmatrix} 1 & 0 \\ 0 & e^{i\alpha} \end{bmatrix} |\psi\rangle = \frac{1}{\sqrt{2}}HP(\alpha)|\psi\rangle. \quad (3.21)$$

For the YZ-plane state, $|m\rangle = c|0\rangle + is|1\rangle$,. Then $\bar{u} = c$, and $\bar{v} = -is$.. Therefore

$$H \begin{bmatrix} \bar{u} & 0 \\ 0 & \bar{v} \end{bmatrix} |\psi\rangle = H \begin{bmatrix} c & 0 \\ 0 & -is \end{bmatrix} |\psi\rangle. \quad (3.22)$$

For the XZ -plane state, $|m\rangle = c|0\rangle + s|1\rangle$,. Hence $\bar{u} = c$, and $\bar{v} = s$. This gives

$$H \begin{bmatrix} \bar{u} & 0 \\ 0 & \bar{v} \end{bmatrix} |\psi\rangle = H \begin{bmatrix} c & 0 \\ 0 & s \end{bmatrix} |\psi\rangle. \quad (3.23)$$

Finally, since H is unitary, $H \begin{bmatrix} \bar{u} & 0 \\ 0 & \bar{v} \end{bmatrix}$ is proportional to a unitary iff $\begin{bmatrix} \bar{u} & 0 \\ 0 & \bar{v} \end{bmatrix}$ is proportional to a unitary. A diagonal matrix $D = \begin{bmatrix} r_0 & 0 \\ 0 & r_1 \end{bmatrix}$ is proportional to a unitary iff

$$D^\dagger D = \begin{bmatrix} |r_0|^2 & 0 \\ 0 & |r_1|^2 \end{bmatrix} = \lambda I \quad (3.24)$$

for some $\lambda > 0$, which holds exactly when $|r_0| = |r_1|$. Thus the XY case is always unitary, while the YZ and XZ cases are unitary only when $|c| = |s|$. $\square$

LEMMA 3.10 The Pauli eigenstates occur as special cases of XY , XZ , and YZ plane measurements:

Pauli effect	Two-plane form	State	ZX effect
$\langle +_X $	$\langle +_{XY,0} $	$\langle + = \frac{1}{\sqrt{2}}(\langle 0 + \langle 1)$	$\bigcirc \text{---}$
$\langle -_X $	$\langle +_{XY,\pi} $	$\langle - = \frac{1}{\sqrt{2}}(\langle 0 - \langle 1)$	$\bigcirc(\pi) \text{---}$
$\langle +_Y $	$\langle +_{XY,\pi/2} $	$\frac{1}{\sqrt{2}}(\langle 0 - i \langle 1)$	$\bigcirc(-\frac{\pi}{2}) \text{---}$
$\langle -_Y $	$\langle +_{XY,-\pi/2} $	$\frac{1}{\sqrt{2}}(\langle 0 + i \langle 1)$	$\bigcirc(\frac{\pi}{2}) \text{---}$
$\langle +_Z $	$\langle +_{XZ,0} = \langle +_{YZ,0} $	$\langle 0 $	$\bullet \text{---}$
$\langle -_Z $	$\langle +_{XZ,\pi} = \langle +_{YZ,\pi} $	$\langle 1 $	$\bullet(\pi) \text{---}$

Consequently, X - and Y -basis measurements are contained in the XY -plane model, while Z -basis measurements require the XZ (or YZ) plane.

Proof. The XY -plane cases follow directly from substituting the relevant angles:

$$|+_{XY,0}\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i0} |1\rangle) = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) = |+_X\rangle, \quad (3.25)$$

$$|+_{XY,\pi}\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi} |1\rangle) = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) = |-_X\rangle. \quad (3.26)$$

Similarly,

$$|+_{XY,\pi/2}\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi/2} |1\rangle) = \frac{1}{\sqrt{2}} (|0\rangle + i |1\rangle) = |+_Y\rangle, \quad (3.27)$$

$$|+_{XY,-\pi/2}\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{-i\pi/2} |1\rangle) = \frac{1}{\sqrt{2}} (|0\rangle - i |1\rangle) = |-_Y\rangle. \quad (3.28)$$

For the Z -basis states, use the XZ plane:

$$|+_{XZ,0}\rangle = \cos(0)|0\rangle + \sin(0)|1\rangle = |0\rangle = |+_Z\rangle, \quad (3.29)$$

$$|+_{XZ,\pi}\rangle = \cos\frac{\pi}{2}|0\rangle + \sin\frac{\pi}{2}|1\rangle = |1\rangle = |-_Z\rangle. \quad (3.30)$$

Finally, every XY -plane state has the form

$$|+_{XY,\alpha}\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{e^{i\alpha}}{\sqrt{2}}|1\rangle. \quad (3.31)$$

Hence both computational-basis amplitudes have magnitude $1/\sqrt{2}$. Therefore no XY -plane state can equal $|0\rangle$ or $|1\rangle$, even up to global phase. Thus Z -basis measurements are not contained in the XY -plane measurement model. $\square$

In this thesis, we are interested in whether there exist measurement patterns on comparability grids that result in universal quantum computation. It remains to formally define what it means to use comparability grids as a resource for universal quantum computation.

3.3 Resource States

DEFINITION 3.11 A **resource state** is a state, typically containing entanglement, on which a measurement pattern can be applied to conduct measurement-based quantum computation. For the purposes of this thesis, a resource state is a graph state.

DEFINITION 3.12 A **family** of resource states is some collection of resource states. A family of resource states Ψ is **universal for MBQC** if, for every n -qubit state $|\phi\rangle$, there exists a resource state $|\Psi\rangle \in \Psi$ on at least n qubits and a measurement pattern $\mathcal{B}$ whose output is $|\phi\rangle$. We say a family of resource states is **n -qubit universal for MBQC** if this holds for all m -qubit $|\phi\rangle$ with $m \leq n$. When the meaning is clear, we may drop the phrase “for MBQC” and simply say a family of resource states is universal.

We start by recalling some well-known results related to graph states.

FACT 3.13 Every graph state has a canonical construction by preparing $|+\rangle^{\otimes |V(G)|}$ and applying one CZ gate for each edge of G , uniquely up to reordering of the commuting CZ gates.

FACT 3.14 Let G be a graph with maximum degree Δ . Ignoring the initial preparation of $|+\rangle^{\otimes |V(G)|}$, $|G\rangle$ can be constructed using at most $\Delta + 1$ layers of CZ gates.

Proof. Vizing’s theorem states that any graph with maximum degree Δ admits an edge coloring with $\Delta + 1$ colors such that no two edges that share a vertex have the same color. All edges of the same color can be implemented in the graph state in the same layer of CZs. Thus, at most $\Delta + 1$ layers are needed. $\square$

FACT 3.15 Given a graph state $|G\rangle$, a Z -basis measurement of a vertex is equivalent to deleting that vertex, up to local Pauli byproducts on its neighbors.

Proof. Consider a graph state $|G\rangle$ in its ZX-Calculus representation.

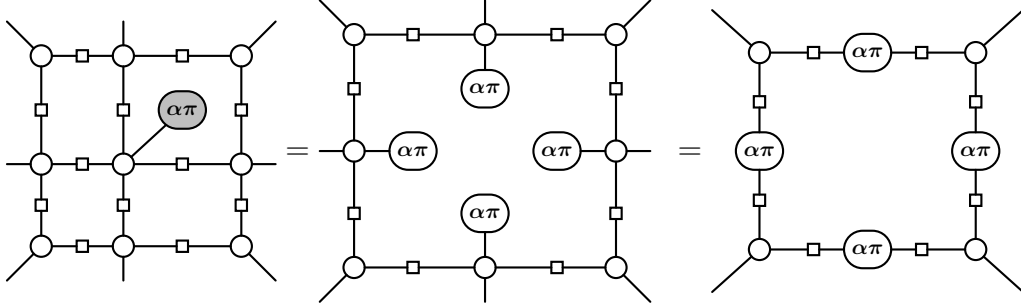


Figure 2: Z plane measurements correspond to vertex deletion.

By Lemma 3.9, the ZX effect of a Z plane measurement is a $\textcircled{a\pi}$ where a is the measurement outcome bit. By (cp), we may push the $\textcircled{a\pi}$ through the vertex being measured, deleting it in the process. The result is a graph state with the measured vertex removed, and a local Z^a byproduct on each neighbor of the measured vertex. $\square$

FACT 3.16 Given a graph state $|G\rangle$, applying $\sqrt{X} = \textcircled{-\pi/2}$ on a vertex v and $\sqrt{Z} = \textcircled{-\pi/2}$ up to global phase on $N_G(v)$ is equivalent to a local complementation around v , and yields $|G \star v\rangle$. Applying a Y plane measurement on v instead gives $|G \star v - v\rangle$ with additional phases.

Proof. The first claim is Lemma 2.28 and the second is Lemma 2.29. $\square$

FACT 3.17 Given a graph state $|G\rangle$ and an edge $uv \in E(G)$, applying H on vertices u, v and Z on $N_G(u) \cap N_G(v)$ gives $|G \wedge uv\rangle$. Applying a X plane measurement on both u and v instead gives $|G \wedge uv - u - v\rangle$ with additional phases.

Proof. The first claim is Lemma 2.31 and the second is Lemma 2.32. $\square$

DEFINITION 3.18 A $m \times n$ grid graph $G = (V, E)$ has

$$V = \{(i, j) \mid i \in [m], j \in [n]\}, \quad (3.32)$$

$$E = \{\{(i_1, j_1), (i_2, j_2)\} \mid (i_1, j_1), (i_2, j_2) \in V, |i_2 - i_1| + |j_2 - j_1| = 1\}. \quad (3.33)$$

In other words, m rows and n columns with edges between nodes of taxicab distance 1. Denote such graphs as $\text{Grid}_{m \times n}$. Let Grid denote the family of all such graphs.

THEOREM 3.19 The Grid is universal.

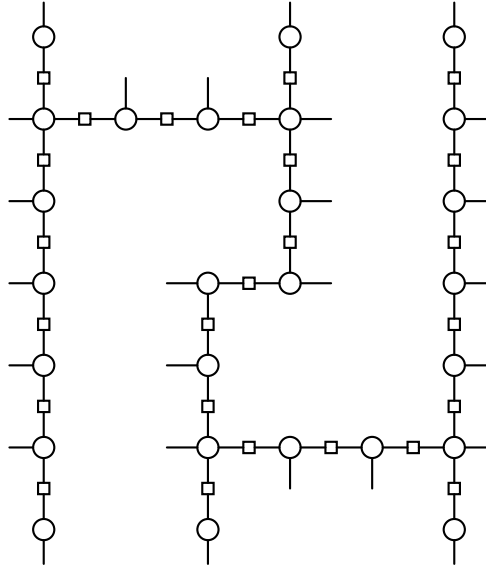


Figure 3: A Grid graph state with some vertices removed, creating three vertical “wires” for a three qubit computation in the MBQC model. The horizontal connections between wires may entangle the qubits, given the right measurements.

Proof. The full proof is quite involved, so we only summarize the main ingredients. The original proof is given in [RB01], and a more accessible proof is available in [RBB03]. A modern proof through the ZX-Calculus is available in Chapter 9 of [KW24].

There are three great difficulties in the proof. We informally discuss them. First, we need to be able to implement arbitrary single-qubit gates. This is done by deleting (via Z plane measurements) vertices to create line graphs, each resulting in a chain like those in Example 3.6. It can be shown that this allows the application of arbitrary single-qubit gates up to some Pauli error.

Second, we need to be able to entangle qubits. If the two line graphs containing the qubits we wish to entangle are adjacent, i.e. there are no other line graphs between them, then an orthogonal connecting chain of vertices can be used to implement an entangling operation between the two logical wires. Now, if they are not adjacent, we can swap qubits between lines (by implementing three CNOT gates) until they become adjacent.

Third, we need to ensure that measurements are performed in the correct order. For example, suppose we delete all vertices connecting the input and output vertices. Then this computation is deterministic and not invertible, and thus does not implement a unitary. As it turns out, there is a criterion called *flow* that ensures this does not happen, as well as variants of flow such as *gflow* and *Pauli flow* that further ensure that we can deal with the Pauli errors. Interested readers should see [MB23; Bac+21; KW24] for more details. $\square$

3.4 Universality Criteria

Given the difficulty of the proof of Theorem 3.19, we now turn our attention to the development of other techniques to determine the universality of a family of resource states. We start by recalling some useful facts regarding regular quantum circuit universality.

FACT 3.20 Any universal set of single-qubit gates, together with any entangling two-qubit gate, is universal [BB01].

Proof. The proof in [BB01] turns out to be quite technical, involving many tricks from Lie algebra. It amounts to proving the following restatement of the theorem: Let V be a 2-qubit gate. The following are equivalent:

1. The collection of all 1-qubit gates $\mathcal{A}$ together with V is QQ-universal.
2. The collection of all 1-qubit gates $\mathcal{A}$ together with V is exactly QQ-universal.
3. V is not primitive, or **imprimitive**.

Here, say V is **primitive** iff $V = S \otimes T$ or $V = (S \otimes T)P$ for some single-qubit gates S and T , with P being the swap gate. $\square$

THEOREM 3.21 Any universal gate set must include at least one multi-qubit entangling gate, as it is impossible to entangle using only single-qubit gates.

Proof. A circuit made only of single-qubit gates maps product states to product states:

$$(U_1 \otimes \cdots \otimes U_n)(|\psi_1\rangle \otimes \cdots \otimes |\psi_n\rangle) = U_1 |\psi_1\rangle \otimes \cdots \otimes U_n |\psi_n\rangle.$$

Hence such a circuit cannot produce an entangled state from a product input. $\square$

We now turn to universality criteria specifically for MBQC. We start by addressing why we only consider CQ-universality rather than QQ-universality.

THEOREM 3.22 This is Observation 2 of [Nes+07]. Every CQ-universal resource can be made QQ-universal by allowing quantum inputs to be coupled into the resource state by attaching the input qubits to the resource using the same entangling operation, namely CZ gates.

We now introduce a new characterization of the universality of graph states.

DEFINITION 3.23 Let **LO** denote local operations, i.e. any single qubit unitary and measurement. Let **LC** denote local Clifford operations, i.e. tensor products of single-qubit Clifford unitaries. Let **CC** denote **classical communication**, which represents the measurement correction and Pauli error tracking performed adaptively in MBQC.

States $|\psi\rangle$ and $|\phi\rangle$ both on n qubits are called **LU-equivalent** if there is a unitary $U = \otimes_{i=1}^n U_i$, where every U_i is a local unitary, such that $|\psi\rangle = U|\phi\rangle$. In this case, write $|\psi\rangle =_{\text{LU}} |\phi\rangle$. Now, if every U_i is a single-qubit Clifford unitary, say they are **LC-equivalent** and write $|\psi\rangle =_{\text{LC}} |\phi\rangle$.

THEOREM 3.24 This is the main result of [VDD04]. Graph states $|G_1\rangle$ and $|G_2\rangle$ are LC-equivalent iff the graphs G_1 and G_2 are related by a sequence of local complementations.

Proof sketch. The reverse direction follows from Lemma 2.28. The forward direction amounts to showing that the successive application of local complementation generates the orbit of any graph state under local Clifford operations. Informally, the proof in [VDD04] does this by a clever characterization of local Clifford operations as graph operations, before showing that all these operations can be implemented using local complementation alone. $\square$

THEOREM 3.25 This is the main result of [Ji+08]. There are graphs that are LU-equivalent but not LC-equivalent. On the other hand, there are some classes of graphs which are LU-equivalent iff they are LC-equivalent [GN07; VDD05].

Proof. In [Ji+08], they present 27-vertex graphs that are LU-equivalent but not LC-equivalent. It was recently shown in [Cla26] that this counterexample is minimal in the number of qubits. $\square$

THEOREM 3.26 A family of states Φ is a universal resource for MBQC iff for every n -qubit state $|\psi\rangle$, there exists $|\phi\rangle \in \Phi$ on $m \geq n$ qubits, such that the transformation $|\phi\rangle \rightarrow |\psi\rangle$ is possible with LO and CC. We denote this by $|\phi\rangle \geq_{\text{LOCC}} |\psi\rangle$.

Proof. This follows almost immediately from the definition. $\square$

DEFINITION 3.27 Let Φ and Ψ be families of resource states. We say that Φ and Ψ are **LU-equivalent**, and write $\Phi =_{\text{LU}} \Psi$, if the following holds:

1. For every $|\phi\rangle \in \Phi$, there exists $|\psi\rangle \in \Psi$ such that $|\phi\rangle =_{\text{LU}} |\psi\rangle$, and
2. for every $|\psi\rangle \in \Psi$, there exists $|\phi\rangle \in \Phi$ such that $|\psi\rangle =_{\text{LU}} |\phi\rangle$.

Define **LC-equivalent** correspondingly.

LEMMA 3.28 Let Φ and Ψ be LU-equivalent families of resource states. Then Φ is a universal resource for MBQC if Ψ is a universal resource for MBQC, and vice versa.

Proof. Suppose Φ is universal, and let $|\rho\rangle$ be any n -qubit target state. By universality, there exists $|\phi\rangle \in \Phi$ on $m \geq n$ qubits such that $|\phi\rangle \geq_{\text{LOCC}} |\rho\rangle$. Since $\Phi =_{\text{LU}} \Psi$, there exists $|\psi\rangle \in \Psi$ such that $|\psi\rangle =_{\text{LU}} |\phi\rangle$. Local unitaries are reversible LOCC operations, so $|\psi\rangle \geq_{\text{LOCC}} |\phi\rangle$. Composing LOCC transformations gives $|\psi\rangle \geq_{\text{LOCC}} |\rho\rangle$. Thus Ψ is universal. The reverse implication is identical. $\square$

Note that, if we have a family of resource states Φ that is not LU equivalent to some universal family of resource states Ψ , this does not necessarily imply Φ is not universal. This concludes our overview of the main universality results concerning LU-, LO, and LC equivalence. To finish off this section, we present one other universality criterion.

THEOREM 3.29 [From [Van+06]] Let Ψ be a family of resource states and $|\phi\rangle, |\phi'\rangle$ arbitrary states with the same number of qubits. Let E be a real-valued function on pure states, possibly taking the value ∞ , satisfying the following properties:

1. $E(|\phi\rangle) \geq E(|\phi'\rangle)$ whenever $|\phi\rangle \geq_{\text{LOCC}} |\phi'\rangle$;
2. $\sup_{\text{Pure states } |\phi\rangle} E(|\phi\rangle) > \sup_{|\psi\rangle \in \Psi} E(|\psi\rangle)$.

Then Ψ cannot be a universal resource.

Proof. The first condition states that **LOCC** cannot increase this measure, while the second says that the supremum value of E , which can be ∞ , is not reached among all $|\psi\rangle \in \Psi$. If Ψ were universal, then for every pure state $|\phi\rangle$ there would be some $|\psi\rangle \in \Psi$ with $|\psi\rangle \geq_{\text{LOCC}} |\phi\rangle$. By monotonicity, $E(|\psi\rangle) \geq E(|\phi\rangle)$. Taking suprema over all target states would give

$$\sup_{|\psi\rangle \in \Psi} E(|\psi\rangle) \geq \sup_{\text{pure states } |\phi\rangle} E(|\phi\rangle), \quad (3.34)$$

contradicting the second condition. $\square$

The above theorem is very general and consequently very powerful, as we will soon see. However, note that satisfying the converse of the second condition for some measure E is not sufficient to guarantee universality. For example, the family of *toric code states* has unbounded entanglement width but are not universal [BR07].

THEOREM 3.30 Consider the family of all star graphs **Star**. A **star graph** on n vertices is a tree where one vertex has degree $n - 1$ and all other vertices have degree 1. Then **Star** is not a universal resource.

We choose to prove a more powerful result using rank-width as our measure E .

DEFINITION 3.31 (RANK-WIDTH) Let $G = (V, E)$ be a graph. For any subset $A \subseteq V$, interpret it as a graph cut with A on one side and $V \setminus A$ on the other. Define the **cut-rank** of A by

$$\rho_G(A) := \text{rank}_{\mathbb{F}_2} M_A, \quad (3.35)$$

where M_A is the $|A| \times |V \setminus A|$ matrix whose entries are

$$(M_A)_{uv} = \begin{cases} 1 & \text{if } u \in A, v \in V \setminus A, \text{ and } uv \in E, \\ 0 & \text{otherwise.} \end{cases} \quad (3.36)$$

In words, $\rho_G(A)$ is the rank, over $\mathbb{F}_2$, of the adjacency matrix recording which edges cross the cut

$$A \mid V \setminus A. \quad (3.37)$$

A **rank decomposition** of G is a binary tree T whose leaves are the vertices of G . Every edge e of T splits the leaves into two parts, and therefore determines a cut

$$A_e \mid V \setminus A_e \quad (3.38)$$

of the vertices of G . The **width** of the decomposition T is

$$\max_{e \in E(T)} \rho_G(A_e). \quad (3.39)$$

Finally, the **rank-width** of G is the smallest possible width over all such decompositions:

$$\text{rwd}(G) := \min_T \max_{e \in E(T)} \rho_G(A_e). \quad (3.40)$$

Thus, rank-width measures how well the vertices of G can be arranged in a tree so that every cut appearing in the tree has low binary adjacency rank.

LEMMA 3.32 Every tree has rank-width at most 1.

Proof. Let $G = (V, E)$ be a tree. We will construct a rank decomposition where every cut has cut-rank at most 1. Root the tree G at some vertex r . For each vertex $v \neq r$, let G_v denote the subtree consisting of v and all of its descendants.

Let $A_v := V(G_v)$. Since G is a tree, there is exactly one edge between A_v and $V \setminus A_v$, namely the edge from v to its parent. Therefore, the cut matrix M_{A_v} has exactly one nonzero entry. Hence

$$\rho_G(A_v) = \text{rank}_{\mathbb{F}_2} M_{A_v} \leq 1. \quad (3.41)$$

Now choose a binary decomposition tree T that recursively separates rooted subtrees of G .⁸ Every cut appearing in this decomposition is of the form $A_v \mid V \setminus A_v$ or a union of such subtrees attached to a common vertex. In either case, because G is a tree, the edges crossing the cut are incident to a single vertex on one side of the cut. Thus the corresponding bipartite adjacency matrix has all nonzero entries contained in a single row or a single column, so its rank over $\mathbb{F}_2$ is at most 1. Hence this rank decomposition has width at most 1. Therefore $\text{rwd}(G) \leq 1$. $\square$

Proof of Theorem 3.30. Every graph in **Star** is a tree, so by Lemma 3.32 any star graph has at most rank-width 1. On the other hand, C_5 , the cycle on 5 vertices, has rank-width at least 2. To see this, take any rank decomposition of C_5 . This tree has 5 leaves, so there must be some edge whose removal separates the leaves into a $2 \mid 3$ split. It suffices to show that every $2 \mid 3$ cut of C_5 has cut-rank 2.

Let C_5 have vertices labelled by $[5]$. Let $A = \{u, v\}$ for some distinct $u, v \in V(C_5)$ and $B = V(C_5) \setminus A$. If u and v are adjacent, by symmetry it suffices to consider $A = \{1, 2\}$. Then

$$M_A = \begin{array}{c|ccc} & 3 & 4 & 5 \\ \hline 1 & 0 & 0 & 1 \\ 2 & 1 & 0 & 0 \end{array}. \quad (3.42)$$

⁸This is always possible. After rooting G , every edge separates a rooted subtree from its complement. Recursively applying such separations gives a decomposition. If a split produces more than two parts, we refine it arbitrarily into binary splits.

Hence the cut-rank of A is 2. Now if u and v are not adjacent, by symmetry it suffices to consider $A = \{1, 3\}$. Then

$$M_A = \begin{array}{c|ccc} & 2 & 4 & 5 \\ \hline 1 & 1 & 0 & 1 \\ 3 & 1 & 1 & 0 \end{array}. \quad (3.43)$$

Thus the cut-rank of A is 2. Hence $\text{rd}(C_5) \geq 2$. Theorem 3.29 completes the proof. $\square$

THEOREM 3.33 Local complementing a graph G at any vertex does not change its rank-width.

Proof. The overall goal is to show that local complementation only performs elementary row and column operations on every cut matrix, so it preserves the cut-rank. For a cut $A \mid B$ of a graph G , let $M_A(G) = M_A$ from Definition 3.31. Then we wish to show

$$\text{rank}_{\mathbb{F}_2} M_A(G) = \text{rank}_{\mathbb{F}_2} M_A(G \star v) \quad (3.44)$$

for all cuts $A \mid B$ and $v \in V(G)$. Suppose first that $v \in A$. Let r_v be the row of $M_A(G)$ indexed by v . Let $u \in A \setminus \{v\}$. Local complementation at v toggles the edge uw , with $w \in B$, exactly when both u and w are adjacent to v . Then the row of $M_A(G \star v)$ indexed by u is

$$r'_u = \begin{cases} r_u + r_v & \text{if } uv \in E(G), \\ r_u & \text{if } uv \notin E(G), \end{cases} \quad (3.45)$$

where the addition is over $\mathbb{F}_2$. The row indexed by v is unchanged, since local complementation does not change edges incident to v . From here, it is not too hard to see that the cut-rank is preserved.

Now suppose $v \in B$. The same argument applies with columns instead of rows. Let c_v be the column of $M_A(G)$ indexed by v . For each $w \in B \setminus \{v\}$, local complementation at v toggles the edge uw , with $u \in A$, exactly when both u and w are adjacent to v . Thus the column indexed by w becomes

$$c'_w = \begin{cases} c_w + c_v & \text{if } vw \in E(G), \\ c_w & \text{if } vw \notin E(G). \end{cases} \quad (3.46)$$

The column indexed by v is unchanged. Once again, it is not too hard to see that the cut-rank is preserved. Therefore every cut has the same cut-rank in G and in $G \star v$. Thus $\text{rd}(G) = \text{rd}(G \star v)$. $\square$

THEOREM 3.34 The family **KGraph** of complete graph states is not universal.

Proof. We first show that every complete graph is local-complementable to a star. Let n be the number of vertices. $n = 1$ is trivial. For $n \geq 2$, local complementation at any $v \in V(K_n)$ toggles all edges not incident to v . The result is a star rooted at v . Thus every complete graph is LC-equivalent to a star graph. Hence the family **KGraph** is LC-equivalent to **Star**. By Lemma 3.28, LC-equivalent families have the same universality status. Since **Star** is not universal by Theorem 3.30, **KGraph** is not universal. $\square$

4 The Comparability Grid

4.1 Circle Graphs

DEFINITION 4.1 The $m \times n$ **comparability grid** is the graph $\text{CGrid}_{m \times n} = (V, E)$ with

$$V = \{(i, j) \mid i \in [m], j \in [n]\}, \quad (4.1)$$

$$E = \{\{(i_1, j_1), (i_2, j_2)\} \mid (i_1, j_1) \neq (i_2, j_2), (i_1 - i_2)(j_1 - j_2) \geq 0\}. \quad (4.2)$$

Equivalently, two distinct vertices are adjacent exactly when they are comparable in the product order on $[m] \times [n]$. Denote this graph by $\text{CGrid}_{m \times n}$. Let CGrid denote the family of all such graphs.

EXAMPLE 4.2 The figure below shows the 3×3 comparability grid:

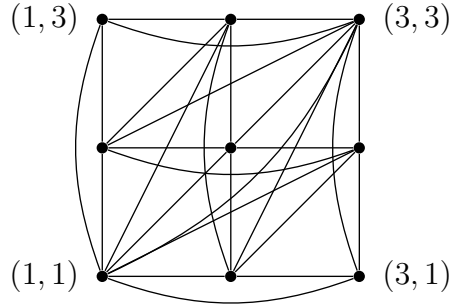


Figure 4: $\text{CGrid}_{3 \times 3}$.

Comparability grids are also circle graphs.

DEFINITION 4.3 A **circle graph** is an intersection graph of chords of a circle. Equivalently, a graph $G = (V, E)$ is a circle graph if we can assign each $v \in V$ a chord c_v in a circle such that

$$uv \in E \iff c_u \text{ and } c_v \text{ intersect.} \quad (4.3)$$

Denote the family of all circle graphs as CGraph .

To see that comparability graphs are circle graphs, we introduce another kind of circle graph: permutation graphs.

DEFINITION 4.4 A **permutation graph** is a graph that records which pairs of objects are in opposite order in two different lists. More precisely, let V be a finite set of labels, and choose two total orders $<_1$ and $<_2$ on V . The permutation graph $G = (V, E)$ associated with these two orders has V as its vertices, and edges

$$uv \in E \iff (u <_1 v \wedge v <_2 u) \vee (v <_1 u \wedge u <_2 v). \quad (4.4)$$

To draw a permutation graph, place the vertices in $<_1$ order on one horizontal line, and place the vertices in $<_2$ order on a second horizontal line below it. Connect matching labels with a line segment. Then two vertices are adjacent exactly when their line segments cross. Denote the family of all permutation graphs as PGraph .

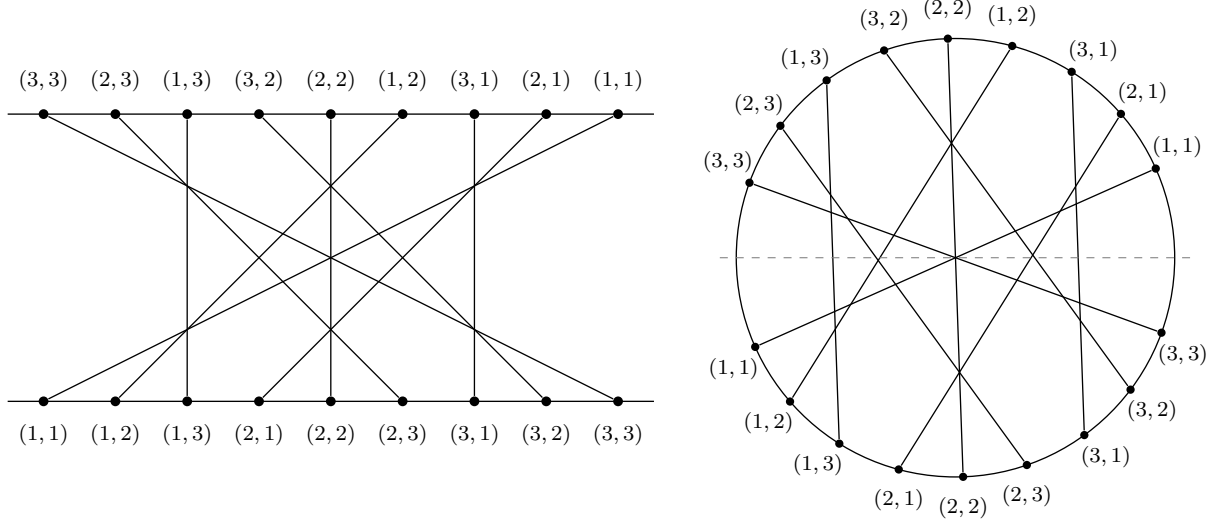


Figure 5: A permutation graph representation of $\mathbf{CGrid}_{3 \times 3}$ and the equivalent circle graph representation.

THEOREM 4.5 A comparability grid is a permutation graph, and hence a circle graph.

Proof. Fix a $G = \mathbf{CGrid}_{m \times n}$. Apply the perturbation from 4.15. This perturbation makes all first coordinates distinct and all second coordinates distinct. Now, define two orders on $V(G)$:

$$(i, j) <_1 (i', j') \iff i < i' \vee (i = i' \wedge j < j') \quad (4.5)$$

$$(i, j) <_2 (i', j') \iff j > j' \vee (j = j' \wedge i > i') \quad (4.6)$$

These are lexicographic orders and thus total orders. Now, draw a permutation diagram with vertices on the top line in order $<_2$ and $<_1$ on the bottom line. Connect equal labels with a line segment. Since the second order is reversed, the two segments cross exactly when the first and second coordinate differences have the same sign. Hence G is a permutation graph.

Finally, we show that any permutation graph is a circle graph. Place the two parallel lines of a permutation graph inside a circle and identify their endpoints along the boundary, such that each line segment becomes a chord of a circle. This preserves crossings. Therefore every permutation graph is a circle graph. $\square$

THEOREM 4.6 This is Theorem 1 of [Hah+26]. For circle graph states, $\mathbf{LU} = \mathbf{LC}$. Furthermore, the only graphs $\mathbf{LU}$ -equivalent to circle graph states are circle graphs themselves. Since local complementation of graph states is implemented by local Clifford operations, and local Clifford operations are local unitaries, this implies that $\mathbf{CGraph}$ is closed under local complementation.

DEFINITION 4.7 A graph H is a **vertex-minor** of a graph G if there is a sequence of vertex deletions and local complementations from G to H .

Proof. By Theorem 4.6, circle graphs are closed under local complementation. They are also closed under vertex deletion, since deleting a vertex from the intersection graph corresponds to deleting the corresponding chord from the chord diagram. $\square$

Proof. This follows from Theorem 4.6. $\square$

THEOREM 4.8 This follows from Corollary 53 of [BT16], together with the strengthening in [Hah+26]. Every circle graph is a vertex-minor of some bipartite circle graph that is at most quadratically larger in the number of vertices.

THEOREM 4.9 This is Lemma 3.1, 3.2, and 3.3 of [Gee+23a]. Every circle graph on n vertices is isomorphic to a vertex-minor of the $3n \times 3n$ comparability grid. More specifically, every n -vertex permutation graph is isomorphic to an induced subgraph of the $n \times n$ comparability grid, and every circle graph on n vertices is a vertex-minor of a permutation graph on $3n$ vertices.

THEOREM 4.10 This follows from Corollaries 3 and 4 of [Hah+26]. MBQC on circle graph states is efficiently classically simulable. Furthermore, assuming $\text{BPP} \neq \text{BQP}^9$, polynomial rank-width is not a sufficient condition for efficient universality of a graph state resources for MBQC.

The previous theorem begins to highlight connections between rank-width and circle graphs. Restricting our attention to comparability grids, we observe even more connections.

THEOREM 4.11 The rank-width of the $n \times n$ comparability grid is at least $n/4$. Thus, CGrid has unbounded rank-width, and so do PGraph and CGraph.

Proof. The first claim is Lemma 4 of [Hah+26]. The second follows by Theorem 4.5. $\square$

THEOREM 4.12 This is Theorem 1.1 and 1.3 of [Gee+23b]. There is a function $f : \mathbb{N} \rightarrow \mathbb{N}$ such that for every positive integer n , every graph of rank-width at least $f(n)$ has a vertex-minor isomorphic to the $n \times n$ comparability grid. On the other hand, there is a function $g : \mathbb{N} \rightarrow \mathbb{N}$ such that for every circle graph H , every graph with no vertex-minor isomorphic to H has rank-width at most $g(|V(H)|)$.

We conclude this section by contrasting the above results for circle graphs and comparability grids with a result for grid graphs.

THEOREM 4.13 This is Theorem 12 and 19 of [Cam+24]. Every planar graph is a vertex-minor of some regular grid. Furthermore, every graph is a vertex-minor of some planar graph. Thus every graph is a vertex-minor of some regular grid.

This is in sharp contrast with the following result.

⁹BPP is the classical complexity class for bounded polynomial-time, defined in [Gil77]. BQP is the quantum complexity class for bounded-error quantum polynomial-time, defined in [BV93]

THEOREM 4.14 This is the main result of [Bou94]. A graph G is a circle graph if and only if no graph locally equivalent to G has an induced subgraph isomorphic to one of the graphs below. These are known as the **forbidden vertex-minors** of circle graphs (and consequently, by Theorem 4.9, of comparability grids).

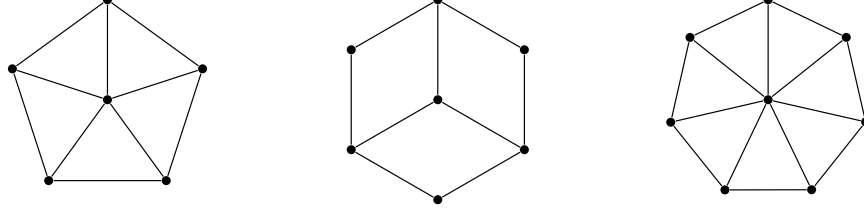


Figure 6: Forbidden vertex minors of circle graphs.

4.2 Subgraphs of the Comparability Grid

Recall that, by Fact 3.15, vertex deletion is an operation allowable in the Measurement Calculus model of MBQC. Thus, if we can show that a family of induced subgraphs of comparability grids is universal, we obtain that comparability grids are universal. Henceforth, when we refer to a subgraph, assume we refer to an induced subgraph unless otherwise noted. We begin by characterizing the data of a subgraph of a comparability grid.

FACT 4.15 Let H be an induced subgraph of $\text{CGrid}_{m \times r}$, with $|V(H)| = n$. Then H admits a canonical encoding by a permutation $\pi(H) \in S_n$. Conversely, every permutation $\pi \in S_n$ determines an induced subgraph $H(\pi)$ of CGrid .

Proof. Write the vertices of H as points (x, y) in the comparability grid, where two vertices are adjacent precisely when they are comparable:

$$(x_1, y_1)(x_2, y_2) \in E(H) \iff (x_1 - x_2)(y_1 - y_2) \geq 0. \quad (4.7)$$

Choose an integer $N > \max\{m, r\}$ and define

$$V' := \left\{ \left(x + \frac{y}{N}, y + \frac{x}{N} \right) : (x, y) \in V(H) \right\}. \quad (4.8)$$

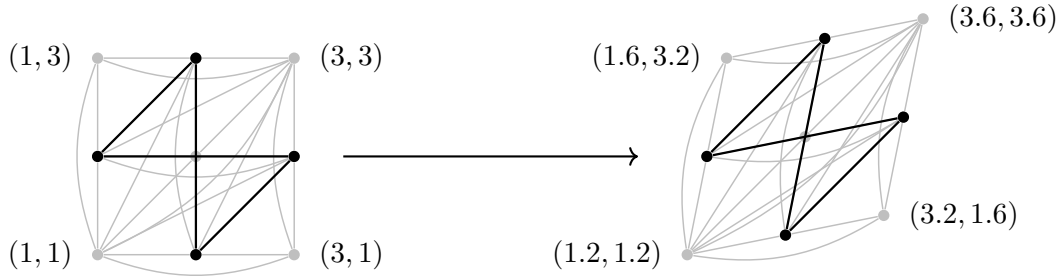


Figure 7: The perturbation of $H(2143)$ in a $\text{CGrid}_{3 \times 3}$.

This perturbation preserves the relative comparability of every pair of vertices, but ensures that no two vertices have the same first coordinate and no two vertices have the same second coordinate. Indeed, if

$$x_1 + \frac{y_1}{N} = x_2 + \frac{y_2}{N} \Rightarrow x_1 - x_2 = \frac{y_2 - y_1}{N}. \quad (4.9)$$

The left-hand side is an integer, while the right-hand side has absolute value strictly less than 1. Hence both sides are 0, so $x_1 = x_2$ and $y_1 = y_2$. Thus the vertices are identical. The same argument applies to the second coordinates.

Now sort the perturbed vertices by increasing first coordinate. Since the second coordinates are all distinct, their relative order is a permutation of $[n]$. Denote this permutation by $\pi(H)$. After relabeling the vertices by increasing perturbed first coordinate, we may represent their relative order abstractly as

$$v_i = (i, \pi(H)(i)), \quad i \in [n]. \quad (4.10)$$

The adjacency rule becomes

$$v_i v_j \in E(H) \iff (i - j)(\pi(H)(i) - \pi(H)(j)) > 0. \quad (4.11)$$

Hence H is the permutation graph encoded by $\pi(H)$, using the convention that the second order is reversed.

Conversely, given $\pi \in S_n$, define

$$V(H(\pi)) := \{(i, \pi(i)) : i \in [n]\}, \quad (4.12)$$

with the induced comparability grid edges. This gives an induced subgraph $H(\pi) \subseteq \text{CGrid}_{n \times n}$. $\square$

FACT 4.16 Two induced subgraphs H, H' of a comparability grid may satisfy $\pi(H) \neq \pi(H')$ while still being isomorphic as abstract graphs.

Proof. Consider the permutations

$$\pi = 132, \quad \sigma = 213. \quad (4.13)$$

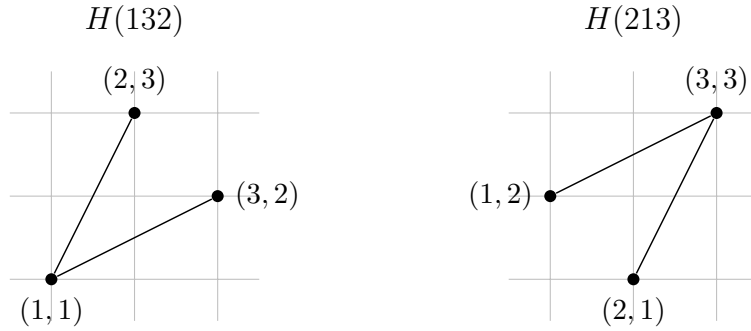


Figure 8: The induced comparability-grid subgraphs $H(132)$ and $H(213)$.

Clearly $H(\pi)$ is isomorphic to $H(\sigma)$ though $\pi \neq \sigma$. $\square$

FACT 4.17 Let $\pi \in S_n$. Define its **reverse-complement** by

$$\pi^{\text{rc}}(i) := n + 1 - \pi(n + 1 - i). \quad (4.14)$$

Then $H(\pi) \cong H(\pi^{\text{rc}})$.

Proof. Write the vertices of $H(\pi)$ as $v_i = (i, \pi(i))$ for $i \in [n]$. Define

$$f(v_i) := (n + 1 - i, n + 1 - \pi(i)). \quad (4.15)$$

As i ranges over $[n]$, the image is exactly

$$\{(k, \pi^{\text{rc}}(k)) : k \in [n]\}, \quad (4.16)$$

so f is a bijection from $V(H(\pi))$ to $V(H(\pi^{\text{rc}}))$.

For $i \neq j$, adjacency in a comparability grid is given by

$$v_i v_j \in E(H(\pi)) \iff (i - j)(\pi(i) - \pi(j)) > 0. \quad (4.17)$$

Under f , both differences change sign:

$$(n + 1 - i) - (n + 1 - j) = -(i - j), \quad (4.18)$$

$$(n + 1 - \pi(i)) - (n + 1 - \pi(j)) = -(\pi(i) - \pi(j)). \quad (4.19)$$

Therefore the product

$$(i - j)(\pi(i) - \pi(j)) \quad (4.20)$$

is preserved. Hence $v_i v_j$ is an edge if and only if $f(v_i) f(v_j)$ is an edge. Thus f is a graph isomorphism, and $H(\pi) \cong H(\pi^{\text{rc}})$. $\square$

This representation of a subgraph will come in handy in Section 5. We now conclude this section with a few observations regarding whether grids are induced subgraphs of some comparability grid.

FACT 4.18 For all $n \in \mathbb{N}$, $\text{Grid}_{2 \times n}$ is an induced subgraph of a $\text{CGrid}_{n+1 \times n+1}$.

Proof. Repeat and “untwist” the following construction: $\square$

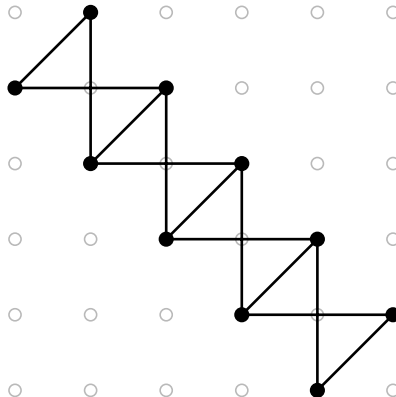


Figure 9: A $\text{Grid}_{2 \times 5}$ in a $\text{CGrid}_{6 \times 6}$.

This result is quite exciting, as it immediately implies that CGrid is 2-qubit universal, as the family of all $\text{Grid}_{2 \times n}$ is 2-qubit universal. Unfortunately, it turns out that we cannot do much better.

FACT 4.19 $\text{Grid}_{3 \times 4}$ is not a vertex-minor of any circle graph. Since vertex-minors are a strictly more powerful operation than vertex deletion, $\text{Grid}_{3 \times 4}$ is not a vertex minor of any circle graph. Since every induced subgraph is a vertex minor, $\text{Grid}_{3 \times 4}$ is not an induced subgraph of any circle graph, and hence not of any comparability grid.

Proof. Look at the following figure:

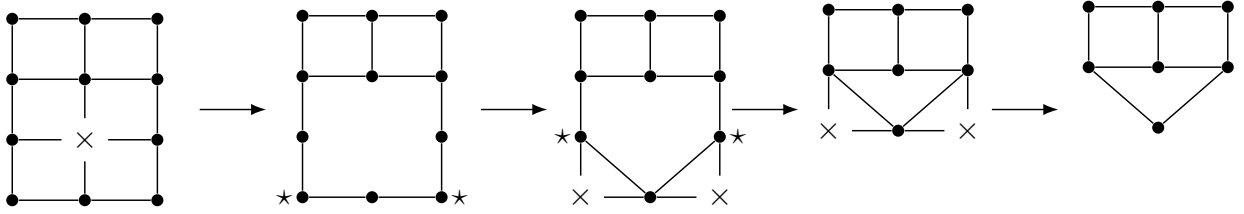


Figure 10: Vertices marked by $\star$ are locally complemented, and vertices marked by $\times$ are deleted. This shows that $\text{Grid}_{3 \times 4}$ has a forbidden minor of circle graphs as a vertex minor.

The rightmost graph is a forbidden vertex-minor of circle graphs. The figure shows that the rightmost forbidden graph is a vertex minor of $\text{Grid}_{3 \times 4}$. If $\text{Grid}_{3 \times 4}$ were a vertex minor of some circle graph, then by transitivity this forbidden graph would also be a vertex minor of a circle graph, contradicting Theorem 4.14. $\square$

Thus, no circle graph, and consequently no comparability grid, has any $\text{Grid}_{m \times n}$ with $m \geq 3$ and $n \geq 4$ as a vertex-minor. An exhaustive search (using the algorithms described in the next section) reveals that we may strengthen this result a little bit.

FACT 4.20 $\text{Grid}_{3 \times 3}$ is not an induced subgraph of any comparability grid. Consequently, $\text{Grid}_{m \times n}$ is not an induced subgraph of any comparability grid for $m, n \geq 3$.

Proof. By Fact 4.15, every embedded 9-vertex induced subgraph of a comparability grid is represented by some permutation in S_9 . Thus it suffices to enumerate the finitely many graphs $H(\pi)$ for $\pi \in S_9$. The enumeration finds no graph isomorphic to $\text{Grid}_{3 \times 3}$. $\square$

Let us summarize what we have seen this section. Through vertex-minors, we see that any potential proof of the universality of the comparability grid cannot involve reducing, via local complementation and vertex deletion, a comparability grid into some desired grid graph. This is the implication of Fact 4.19.

Furthermore, Theorem 4.10 rules out an efficient MBQC universality proof for comparability grids under the assumption $\text{BPP} \neq \text{BQP}$. In particular, an efficient construction using only polynomial-size circle-graph resources and polynomial-time classical control would yield an efficient classical simulation, contradicting the expected separation between BPP and BQP .

Suppose comparability grids are universal. If we are to present a constructive proof of this, i.e. given a target state $|\phi\rangle$ on n qubits, an explicit vertex-minor of some comparability grid

and a measurement pattern that generates $|\phi\rangle$ using it, then this vertex-minor would have to use super-polynomially many vertices with respect to n . In the next section, we outline the key methodology of this thesis: a series of algorithms for enumerating and characterizing the vertex-minors of comparability grids. We use these algorithms to find constructions that we conjecture may lead to a proof of the universality of the comparability grids.

5 Algorithmic Approaches

5.1 Motivation

We begin by reviewing our problem statement. We wish to determine if **CGrid** is universal. If this is true, then by Theorem 4.9, the family of all circle graph states is universal. Moreover, by Theorem 4.12, the same would hold for any graph-state family to which that vertex-minor reduction applies. On the other hand, assuming $\text{BQP} \neq \text{BPP}$, Theorem 4.10 tells us that such universality cannot be efficient. Furthermore, if the comparability grid is not universal, unbounded rank-width is a necessary (by Theorem 3.29) but not sufficient condition for universality (strengthening a result from [Hah+26]).

We saw in Section 3 why this problem may not be as simple as it first seems. Notably, we saw how **KGraph** and any family consisting only of tree graph states cannot be universal for quantum computation. Informally, this tells us that both too few and too many edges may be problematic for universality. On the other hand, **Grid** is universal. However, through forbidden minors we see that any $\text{Grid}_{3 \times 4}$ or larger cannot be a vertex-minor of any **CGrid**. Thus, a proof of the universality of **CGrid** cannot involve a reduction to the universality of **Grid**.

In this section, we introduce algorithmic tools that give evidence for the universality of comparability grids. These algorithmic tools can be divided into three parts, each addressing the following questions:

1. Section 5.2: Subgraph Algorithms. Can we conduct a census of small-size vertex-minors of **CGrid**? What patterns emerge? What kinds of patterns may be advantageous for MBQC?
2. Section 5.3: Circuit Extraction. Given a subgraph of the comparability grid, what kind of quantum computations can we conduct on them through MBQC? Can we characterize this computational power through a set of quantum gates?
3. Section 5.4: Universality Computations. Given a gate set, can we computationally determine whether it is universal?

5.2 Subgraph Algorithms

FACT 5.1 Given a finite graph $G = (V, E)$, we can decide whether it is a vertex-minor of some circle graph in linear time, with respect to the number of vertices and edges of G .

Proof. By Theorem 4.6, circle graphs are closed under vertex-minors. Thus to decide whether G is a vertex-minor of some circle graph is equivalent to deciding whether G is a circle graph. [PR25] presents a linear time algorithm for this, using PC-trees. $\square$

Note that by Theorem 4.9, the above is equivalent to deciding if G is a vertex-minor of some comparability grid. If we restrict our attention to deciding if G is an induced subgraph of some comparability grid, we obtain similar results.

FACT 5.2 A finite graph G is an induced subgraph of a comparability grid if and only if G is a permutation graph. Furthermore, the data of a k -vertex permutation graph obtained in this way is completely characterized by a permutation $\pi \in S_k$.

Proof. This is a restatement of Fact 4.15. □

LEMMA 5.3 Let R_k be the number of permutations of $[k]$, modulo the equivalence relation generated by reverse-complement $\pi^{rc}(i) := k + 1 - \pi(k + 1 - i)$. Then

$$R_k = \frac{k! + 2^{\lfloor k/2 \rfloor} \lfloor k/2 \rfloor!}{2}. \quad (5.1)$$

Proof. The operation $\pi \mapsto \pi^{rc}$ has order 2, so each equivalence class has size 1 or 2. By Burnside's lemma, the number of equivalence classes is the average number of permutations π fixed by the two operations: the identity operation and reverse-complement. Hence

$$R_k = \frac{\#\{\pi \text{ fixed by the identity}\} + \#\{\pi \text{ fixed by reverse-complement}\}}{2}. \quad (5.2)$$

The identity fixes every permutation, so the first term is $k!$.

It remains to count the permutations fixed by reverse-complement. Such a permutation satisfies $\pi = \pi^{rc}$, which means

$$\pi(i) = k + 1 - \pi(k + 1 - i). \quad (5.3)$$

Equivalently, $\pi(i) + \pi(k + 1 - i) = k + 1$. Thus the positions come in symmetric pairs $\{1, k\}, \{2, k - 1\}, \dots$, and the values also come in symmetric pairs $\{1, k\}, \{2, k - 1\}, \dots$. There are $\lfloor k/2 \rfloor$ such position-pairs and $\lfloor k/2 \rfloor$ such value-pairs.

To build a reverse-complement-fixed permutation, first match the position-pairs with the value-pairs. This can be done in $\lfloor k/2 \rfloor!$ ways. Then, for each matched pair, choose which of the two values goes in the smaller position. This gives $2^{\lfloor k/2 \rfloor}$ choices. If k is odd, the middle position is forced to take the middle value, so it gives no extra choice. Therefore the number of permutations fixed by reverse-complement is

$$2^{\lfloor k/2 \rfloor} \lfloor k/2 \rfloor!. \quad (5.4)$$

Substituting into Burnside's lemma gives the desired result. □

We remark that, for $k \geq 3$, R_k is strictly smaller than $k!$. According to the enumeration recorded as OEIS A156808, for $k \geq 10$ R_k is smaller than the number of connected circle graphs on k vertices [OEI].

We would like this number to be as small as possible so that we may more easily conduct a census of all k -vertex subgraphs of CGrid. For example, we may be interested in a census of the degree sequences of all k -vertex subgraphs of CGrid. The following fact and lemma show that we may do this in $O(R_k \cdot k \log k)$ time.

FACT 5.4 In order to enumerate every k -vertex subgraph of any CGrid up to graph isomorphism, it suffices to enumerate R_k objects.

Proof. This follows from Fact 5.2 and Lemma 5.3. $\square$

LEMMA 5.5 Let $\pi \in S_k$. We can compute the degree sequence of $H(\pi)$ in $O(k \log k)$ time.

Proof. Recall that $ij \in E(H(\pi))$ iff

$$(i - j)(\pi(i) - \pi(j)) \geq 0. \quad (5.5)$$

Thus the degree of vertex i splits into a **forward contribution** and a **backward contribution**:

$$\deg(i) = \#\{j < i : \pi(j) < \pi(i)\} + \#\{j > i : \pi(i) < \pi(j)\}. \quad (5.6)$$

A Fenwick tree is a well-known data structure that stores counts over the values $[k]$. It supports two operations in $O(\log k)$ time: inserting a value t , and querying how many inserted values are at most t . Thus, after inserting some subset of values, a query at t returns

$$\#\{\text{inserted values} \leq t\}. \quad (5.7)$$

We compute the **forward term** by scanning π left to right with a Fenwick tree. At step i , the tree stores the values $\pi(1), \dots, \pi(i-1)$. Therefore a query at $\pi(i) - 1$ gives

$$L_i = \#\{j < i : \pi(j) < \pi(i)\}. \quad (5.8)$$

Then insert $\pi(i)$. The **backward term** is analogous, but we scan from right to left. Each scan performs $O(k)$ Fenwick-tree updates and queries, each taking $O(\log k)$ time. Hence the total runtime is $O(k \log k)$. $\square$

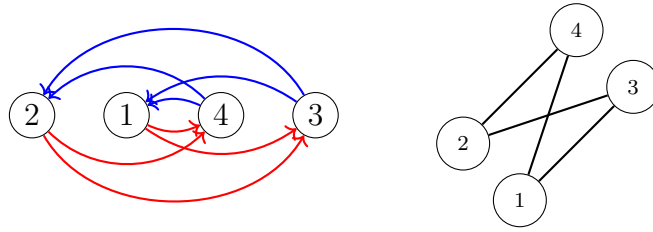


Figure 11: For $\pi = 2143$, the left figure shows the **forward** and **backward** contributions to the degree sequence. The right figure shows $H(\pi)$.

With these results in hand, we can conduct a census of k -vertex subgraphs relatively easily. For example, we can enumerate all 9-vertex subgraphs of **CGrid** that themselves have the $\text{Grid}_{3 \times 3}$ as an induced subgraph. From this, we obtain that the $\text{Grid}_{3 \times 3}$ is not an induced subgraph of any comparability grid (Fact 4.20). On the other hand, we obtain two interesting induced subgraphs. One has two additional edges compared to $\text{Grid}_{3 \times 3}$, while the other has five. We will introduce these subgraphs as tileable graph gadgets.

DEFINITION 5.6 A **graph gadget** is a tuple

$$\Lambda = (H, L, R), \quad (5.9)$$

where H is a finite induced subgraph of $\mathbf{CGrid}$, and

$$L = (\ell_1, \dots, \ell_n), \quad R = (r_1, \dots, r_n), \quad (5.10)$$

are ordered subsets of $V(H)$ called the **left boundary** and **right boundary**. We assume $L \cap R = \emptyset$. We may also call Λ a **n -wire graph gadget**.

DEFINITION 5.7 Let $\Lambda = (H, L, R)$ be an n -wire graph gadget, with $L = (\ell_1, \dots, \ell_n)$ and $R = (r_1, \dots, r_n)$. Say that Λ is **tileable** if, for every positive integer a , there exist induced subgraphs

$$H^{(1)}, H^{(2)}, \dots, H^{(a)} \subseteq \mathbf{CGrid} \quad (5.11)$$

such that each $H^{(m)}$ is isomorphic to H . Let $L^{(m)} = (\ell_1^{(m)}, \dots, \ell_n^{(m)})$ and $R^{(m)} = (r_1^{(m)}, \dots, r_n^{(m)})$ denote the corresponding left and right boundary vertices of $H^{(m)}$. We require that consecutive copies are glued boundary-to-boundary:

$$r_j^{(m)} = \ell_j^{(m+1)} \quad \text{for every } m \in [a-1] \text{ and } j \in [n]. \quad (5.12)$$

Furthermore, the union of these glued copies must be an induced subgraph of $\mathbf{CGrid}$ with no edges except those already appearing inside the individual copies.

Equivalently, Λ is tileable if arbitrarily many copies of H can be embedded in a comparability grid in a chain, with the right boundary of each copy glued to the left boundary of the next, without introducing any additional edges.

DEFINITION 5.8 Let the **diagonal graph gadget** denote the induced subgraph of $\mathbf{CGrid}$ given by the permutation $\pi = 649281753$, with left boundary $L = (9, 2, 5)$ and right boundary $R = (8, 4, 3)$.

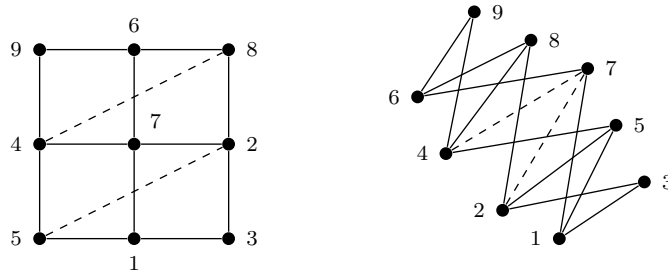


Figure 12: The diagonal graph viewed as a grid with additional edges (left) and as a subgraph of $\mathbf{CGrid}$ (right).

FACT 5.9 The diagonal gadget is tileable.

Proof. Look at the following figure. □

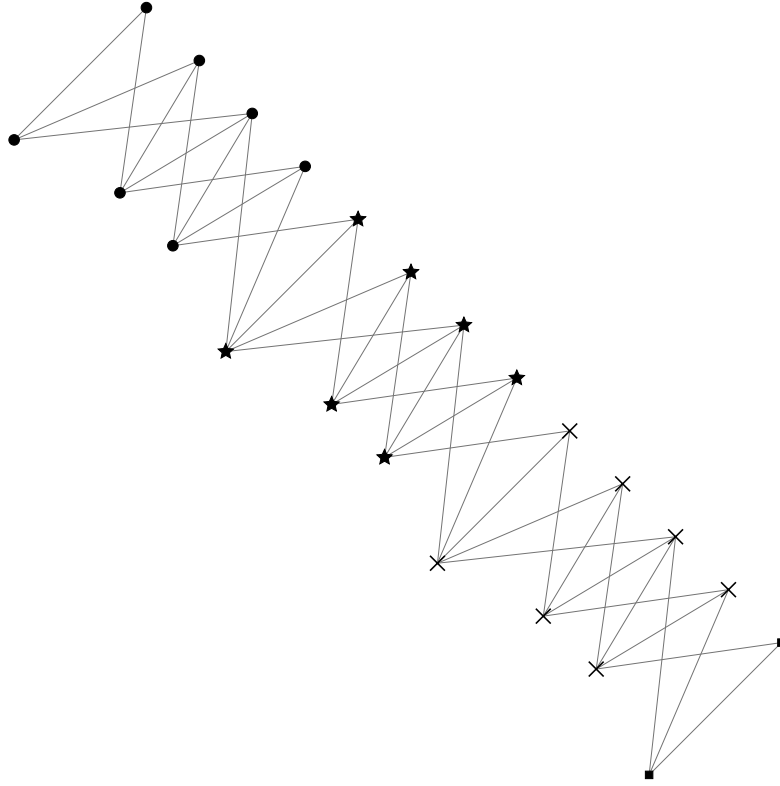


Figure 13: Three tiled copies of the diagonal graph gadget. The first copy excluding its right boundary is shown in $\bullet$. The second copy excluding its right boundary is shown in $\star$. The third copy excluding its right boundary is shown in $\times$. The third copy's right boundary is shown in $\blacksquare$.

DEFINITION 5.10 Let the **width 3 gadget** denote the induced subgraph of CGrid given by the permutation $\pi = 628491735$, with left boundary $L = (9, 6, 8)$ and right boundary $R = (5, 1, 3)$.

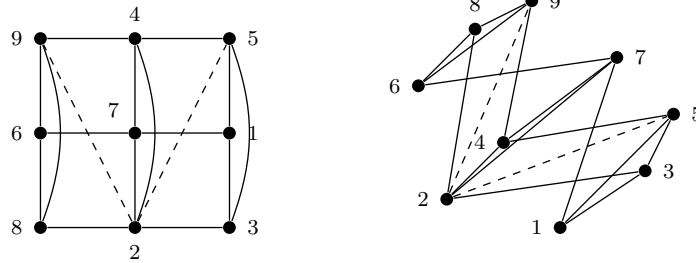


Figure 14: The width 3 graph gadget viewed as a grid with additional edges (left) and as a subgraph of CGrid (right).

Proof. Look at the following figure. □

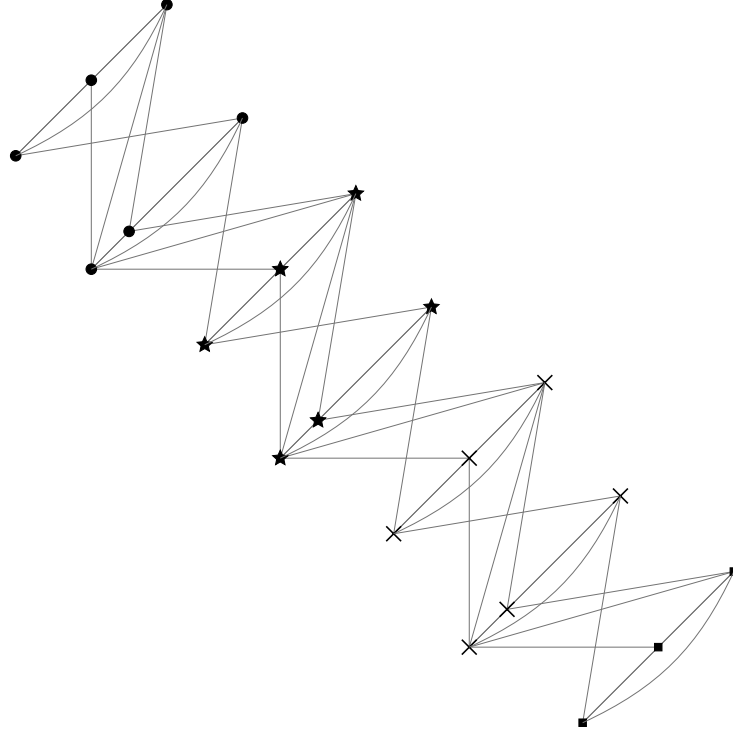


Figure 15: Three tiled copies of the width 3 graph gadget. The first copy excluding its right boundary is shown in •. The second copy excluding its right boundary is shown in ★. The third copy excluding its right boundary is shown in ×. The third copy's right boundary is shown in ■.

The previous two proofs are visual proofs purely for brevity. Indeed, we can formally verify these visual proofs by the following lemma.

LEMMA 5.11 (TILEABLE WITNESS) Let $\Lambda = (H, L, R)$ be an n -wire graph gadget. Suppose we are given a vector $t = (\Delta x, \Delta y)$ with $\Delta x > 0$ and $\Delta y < 0$ such that $r_j = \ell_j + t$ for every $j \in [n]$. Let

$$X := \max_{p,q \in V(H)} |x(p) - x(q)|, \quad Y := \max_{p,q \in V(H)} |y(p) - y(q)|, \quad (5.13)$$

and set

$$A := 2 + \max \left\{ \left\lceil \frac{X}{\Delta x} \right\rceil, \left\lceil \frac{Y}{-\Delta y} \right\rceil \right\}. \quad (5.14)$$

Determine an embedding of H into the CGrid. If the union of the first A translated copies, with the intended boundary identifications, has no edges except those contained in the copies themselves, then Λ is tileable.

Proof. Suppose the first A translated copies have no extra edges. We show that no later copy can introduce one. Let $p, q \in V(H)$, and consider vertices p and $q + dt$ from two copies separated by $d \geq 1$ steps. An edge exists in **CGrid** exactly when

$$q + dt - p = (x, y) \quad (5.15)$$

has both coordinates nonnegative or both coordinates nonpositive. Since $t = (\Delta x, \Delta y)$ with $\Delta x > 0$ and $\Delta y < 0$, increasing d increases the first coordinate and decreases the second. If

$$d > \max \left\{ \left\lceil \frac{X}{\Delta x} \right\rceil, \left\lceil \frac{Y}{-\Delta y} \right\rceil \right\}, \quad (5.16)$$

then the first coordinate of $q + dt - p$ is positive and the second coordinate is negative. Hence the two vertices are incomparable, so no edge can appear. Therefore every possible extra edge would already occur between two of the first A copies. Since no such edge occurs there, no extra edge occurs in any longer chain. Thus arbitrarily many translated copies may be glued in sequence, so Λ is tileable. Finally, note that this check runs in time roughly $O(A^2|V(H)|^2)$. $\square$

Without this witness embedding and translation vector t , it does not seem we can efficiently decide if a given graph gadget is tileable.

DEFINITION 5.12 Let $\Lambda = (H, L, R)$ be an n -wire graph gadget, and let $a \geq 1$. The **a -chain graph** of Λ , denoted $T_a(\Lambda)$, is obtained from a disjoint labeled copies

$$H^{(1)}, H^{(2)}, \dots, H^{(a)}$$

by identifying the right boundary of each copy with the left boundary of the next copy:

$$r_j^{(m)} \sim \ell_j^{(m+1)} \quad \text{for every } m \in [a-1] \text{ and } j \in [n]. \quad (5.17)$$

Equivalently, $T_a(\Lambda)$ is the abstract graph obtained by gluing copies of H boundary-to-boundary in a chain.

DEFINITION 5.13 Let $\Lambda = (H, L, R)$ be an n -wire graph gadget. Say that Λ is **a -tileable** if there exists a point set $P \subseteq \mathbb{Z}^2$ such that

$$\mathbf{CGrid}[P] \cong T_a(\Lambda). \quad (5.18)$$

Say that Λ is **tileable** if it is a -tileable for every positive integer a .

LEMMA 5.14 Let $P \subseteq \mathbb{Z}^2$ be a finite point set with $|P| = N$. Then there exists $P' \subseteq [N] \times [N]$ such that

$$\mathbf{CGrid}[P'] \cong \mathbf{CGrid}[P]. \quad (5.19)$$

Proof. Replace each x -coordinate by its rank among the distinct x -coordinates appearing in P , and do the same for y -coordinates. Thus $p = (x(p), y(p))$ is sent to

$$p' = (\text{rank}_x(x(p)), \text{rank}_y(y(p))). \quad (5.20)$$

This preserves the coordinate-wise order. Hence, for all $p, q \in P$,

$$x(p) \leq x(q) \text{ and } y(p) \leq y(q) \iff x(p') \leq x(q') \text{ and } y(p') \leq y(q'). \quad (5.21)$$

Therefore comparability edges are preserved, so $\text{CGrid}[P'] \cong \text{CGrid}[P]$. Since at most N distinct ranks occur in each coordinate, $P' \subseteq [N] \times [N]$. $\square$

Algorithm 1 Finite test for a -tileability

Input: An n -wire graph gadget $\Lambda = (H, L, R)$ and $a \geq 1$.

Construct the glued chain graph $T_a = T_a(\Lambda)$.

Let $N = |V(T_a)| = a(|V(H)| - n) + n$.

Enumerate functions

$$X : V(T_a) \rightarrow [N], \quad Y : V(T_a) \rightarrow [N]. \quad (5.22)$$

For each pair (X, Y) , first check that the map

$$v \mapsto (X(v), Y(v)) \quad (5.23)$$

is injective. If it is not injective, discard (X, Y) .

Otherwise, check whether for every distinct $u, v \in V(T_a)$,

$$uv \in E(T_a) \iff (X(u) \leq X(v) \text{ and } Y(u) \leq Y(v)) \text{ or } (X(v) \leq X(u) \text{ and } Y(v) \leq Y(u)). \quad (5.24)$$

If this condition holds for all pairs u, v , accept.

If no coordinate assignment passes, reject.

THEOREM 5.15 Algorithm 1 accepts if and only if Λ is a -tileable.

Proof. Suppose the algorithm accepts. Then there are functions $X, Y : V(T_a) \rightarrow [N]$ such that $v \mapsto (X(v), Y(v))$ is injective and preserves exactly the comparability edges. Let

$$P := \{(X(v), Y(v)) : v \in V(T_a)\}. \quad (5.25)$$

Then $P \subseteq [N] \times [N] \subseteq \mathbb{Z}^2$, and the pairwise check says precisely that

$$\text{CGrid}[P] \cong T_a. \quad (5.26)$$

Hence Λ is a -tileable.

Conversely, suppose Λ is a -tileable. Then there exists a point set $P \subseteq \mathbb{Z}^2$ such that $\text{CGrid}[P] \cong T_a$. Since $|V(T_a)| = N$, Lemma 5.14 gives an isomorphic realization $P' \subseteq [N] \times [N]$. Writing each vertex $v \in V(T_a)$ as $v \mapsto (X(v), Y(v)) \in P'$, the algorithm will eventually enumerate this coordinate assignment and accept. $\square$

The algorithm enumerates $N^{2N} = e^{O(N \log N)}$ coordinate assignments, and each assignment can be checked in $O(N^2)$ time. Hence the total runtime is $e^{O(N \log N)}$.

5.3 Circuit Extraction

In the previous section, we identified two tileable graph gadgets: the diagonal gadget and the width 3 gadget. In this section, we see how we can associate the MBQC computational power of a tileable graph gadget with a single gate.

DEFINITION 5.16 Let $\Lambda = (H, L, R)$ be an n -wire graph gadget, with $L = (\ell_1, \dots, \ell_n)$ and $R = (r_1, \dots, r_n)$. The **open tile** associated to Λ is the induced subgraph

$$H^\circ := H[V(H) \setminus R]. \quad (5.27)$$

We call L the **input boundary** of H° . The right boundary R is removed because, when copies of Λ are tiled, the right boundary of one copy is identified with the left boundary of the next.

DEFINITION 5.17 Let $\Lambda = (H, L, R)$ be an n -wire graph gadget. A **measurement pattern** on Λ is a function

$$\mu : V(H^\circ) \setminus L \rightarrow \{\text{measurement planes}\} \times \{\text{angle parameters}\}. \quad (5.28)$$

We write θ for the tuple of all free angles. We also set the convention, for computational reasons, that measurements are conducted one “column” at a time, starting from L , then proceeding to the non-measured vertices adjacent to L , and so on.¹⁰

LEMMA 5.18 For the unitary gadget gates considered in this section, after the graph gadget has already been fixed, it suffices to consider XY -plane measurements.

Proof. By Lemma 3.9, the unnormalized one-step effect of a unitary single-qubit measurement gadget must be proportional to

$$H \begin{bmatrix} u & 0 \\ 0 & v \end{bmatrix} \propto HP(\theta) \quad (5.29)$$

for some $\theta \in [0, 2\pi)$. Thus, as we are not conducting any vertex deletion, it suffices to consider XY plane measurements and μ is now a map to angle parameters alone. $\square$

DEFINITION 5.19 Let $\Lambda = (H, L, R)$ be an n -wire graph gadget and let μ be a measurement pattern on Λ . The **gadget gate** associated to (Λ, μ) is the linear map

$$G_{\Lambda, \mu}(\theta) : (\mathbb{C}^2)^{\otimes n} \rightarrow (\mathbb{C}^2)^{\otimes n} \quad (5.30)$$

that implements the following: Place the input state on the ordered vertices $L = (\ell_1, \dots, \ell_n)$, prepare every vertex in $V(H^\circ) \setminus L$ in $|+\rangle$, apply CZ along every edge of H° , and measure the vertices in $V(H^\circ) \setminus L$ according to μ . After applying the Pauli byproduct corrections determined by the measurement outcomes, if the resulting map is proportional to a unitary for all choices of θ , we call (Λ, μ) a unitary gadget.

¹⁰One must also ensure that some form of *flow* exists for this to be a valid quantum operation, as remarked in the proof of Theorem 3.19. We take this for granted for now, to simplify computations.

Now, how can we find the gadget gate of a graph gadget? Let us compute a simple example together.

FACT 5.20

$$\begin{array}{c} q_0 \\ q_1 \\ q_2 \end{array} \begin{array}{c} \bullet \\ \bullet \\ \oplus \end{array} \begin{array}{c} \oplus \\ \bullet \\ \oplus \end{array} = \begin{array}{c} q_0 \\ q_1 \\ q_2 \end{array} \begin{array}{c} \oplus \\ \bullet \\ \oplus \end{array} \begin{array}{c} \bullet \\ \oplus \\ \oplus \end{array} \quad (5.31)$$

Proof. Note that

$$\begin{array}{c} \circ \\ \bullet \\ \bullet \end{array} \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} \xrightarrow{(b)} \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} \begin{array}{c} \circ \\ \circ \\ \circ \end{array} \xrightarrow{(f)} \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} \begin{array}{c} \circ \\ \circ \\ \circ \end{array} \xrightarrow{(f)} \begin{array}{c} \bullet \\ \bullet \\ \bullet \end{array} \begin{array}{c} \circ \\ \circ \\ \circ \end{array} \quad (5.32)$$

This completes the proof. $\square$

EXAMPLE 5.21 We compute the gadget gate of the diagonal gadget. Here is our starting graph state and XY plane measurements:

$$\begin{array}{c} |\psi\rangle \\ |+\rangle \\ |+\rangle \\ |+\rangle \end{array} \begin{array}{c} \bullet \\ \bullet \\ \bullet \\ \bullet \end{array} \begin{array}{c} \bullet \\ \bullet \\ \bullet \\ \bullet \end{array} \begin{array}{c} R_Z(\theta_1) \\ R_Z(\theta_2) \\ R_Z(\theta_3) \\ \text{---} \end{array} \begin{array}{c} H \\ H \\ H \\ \text{---} \end{array} \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \quad (5.33)$$

Pushing gates through like in Equation 3.2, we get

$$\begin{array}{c} (H^{\otimes 3} R_Z(\vec{\theta})) |\psi\rangle \\ |+\rangle \\ |+\rangle \\ |+\rangle \end{array} \begin{array}{c} \oplus \\ \oplus \\ \oplus \\ \oplus \end{array} \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \quad (5.34)$$

We can add the following CNOTs without affecting the circuit, like in Equation 3.3:

Quantum circuit diagram (5.35) showing a sequence of gates and measurements on 7 qubits. The top two qubits are initialized to $(H^{\otimes 3} R_Z(\vec{\theta})) |\psi\rangle$. The bottom three qubits are initialized to $|+\rangle$. The circuit consists of a series of controlled operations (CNOTs and controlled rotations) and measurements on the top three qubits.

Applying the swap gate identity from Equation 3.4, we get

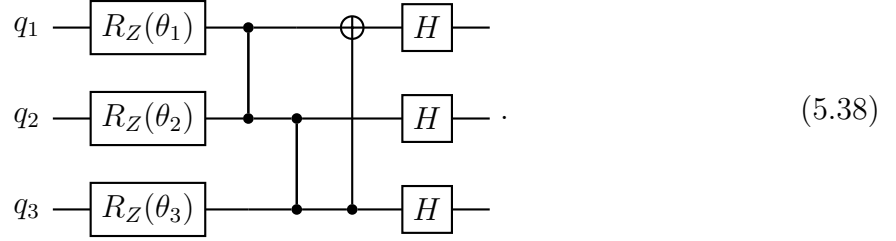
$$\begin{array}{c}
 \left. \begin{array}{l}
 \text{---} \times \text{---} \bullet \text{---} \boxed{\nearrow} \\
 \text{---} \times \text{---} \bullet \text{---} \boxed{\nearrow} \\
 \text{---} \times \text{---} \oplus \text{---} \bullet \text{---} \boxed{\nearrow}
 \end{array} \right\} (H^{\otimes 3} R_Z(\vec{\theta})) |\psi\rangle \\
 \begin{array}{l}
 |+\rangle \text{---} \times \text{---} \oplus \text{---} \bullet \text{---} \text{---} \bullet \text{---} \\
 |+\rangle \text{---} \times \text{---} \oplus \text{---} \times \text{---} \oplus \text{---} \bullet \text{---} \text{---} \bullet \text{---} \\
 |+\rangle \text{---} \times \text{---} \oplus \text{---} \oplus \text{---} \oplus \text{---}
 \end{array}
 \end{array} \quad (5.36)$$

Relabeling wires by moving the swaps to the beginning gives

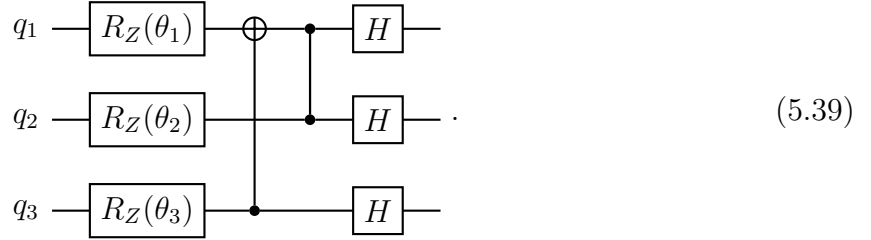
(5.37)

Tracking the Pauli byproducts classically and suppressing them from the diagram, the non-

Pauli part of the implemented block is

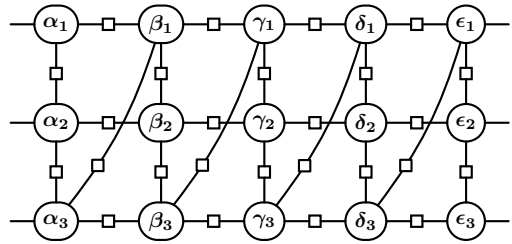


Call the above the **diagonal gadget gate**. Using Fact 5.20, the gadget is equivalent to

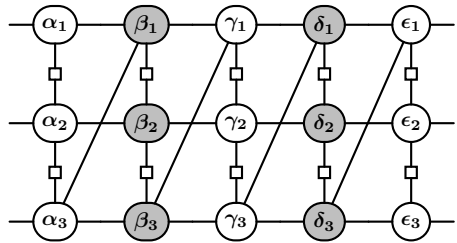


Now, let us do the same procedure in the ZX calculus.

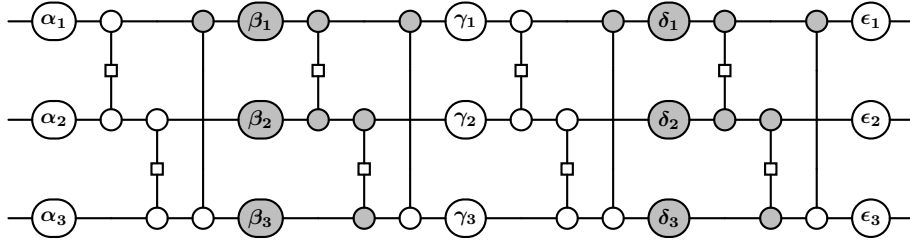
EXAMPLE 5.22 Since $(a\pi) \square \square \pi = ((a+1)\pi)$, we may write 4 open tiles with one additional column of the diagonal gadget gate with measurements as below:



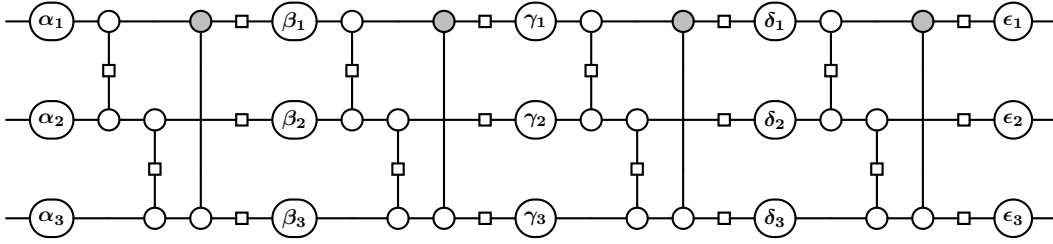
Applying (cc) gives us



Applying (f) gives



Finally, one more application of (cc) brings us to the same diagonal gadget gate:



THEOREM 5.23 Both the quantum circuit and ZX Calculus circuit extraction methods can be automated.

Proof. The algorithm is not very insightful and amounts to the bookkeeping of many cases, so we spare readers the details. Much of the key insights of this algorithm can be found in the proof of Theorem 2.19. $\square$

Thus every extracted gadget gate has the form $R_Z(\theta) \circ C$ for a Clifford circuit C . This turns out to be extremely useful.

5.4 Lie Algebra Computations

In this section, we recall a well-known Lie-algebraic method for showing the approximate universality of a gate set in a purely computational manner. Here is a high level overview of the approach: Using Lie-theoretic arguments, it can be shown that given a gate set, it is approximately universal if and only if it generates the Lie algebra of the special unitary group $SU(2^n)$. It turns out that we get the operators $-iZ_1, \dots, -iZ_n$ for “free” as generators given our gadget gate definition. Now, we have three ways of generating new operators from the above: conjugation by the Clifford circuit attached to the gadget, addition, and commutation. All three of these operations can be done efficiently via the tableau representation in Definition 2.45, with respect to the number of qubits of the gadget. Therefore, we can compute the Lie algebra generated by the gate set, and if it is the full Lie algebra, then we have computationally verified that the gate set is approximately universal.

Note that for simplicity, we ignore global phases throughout. In other words, we work inside $SU(2^n)$ rather than $U(2^n)$.

DEFINITION 5.24 The **n -qubit Clifford group** is the normalizer of the Pauli group in the unitary group:

$$\mathcal{C}_n := \{C \in U(2^n) \mid C\mathcal{P}_n C^\dagger = \mathcal{P}_n\}. \quad (5.40)$$

Equivalently, $C \in \mathcal{C}_n$ if and only if conjugation by C sends Pauli operators to Pauli operators.

DEFINITION 5.25 For any $A \in \mathbb{C}^{N \times N}$, define the **matrix exponential** of A by

$$e^A := \sum_{k=0}^{\infty} \frac{A^k}{k!}. \quad (5.41)$$

DEFINITION 5.26 Let $A \in \mathbb{C}^{n \times n}$. We say that A is **Hermitian** if $A^\dagger = A$, **skew-Hermitian** if $A^\dagger = -A$, and **traceless** if $\text{tr}(A) = 0$, where $\text{tr}(A)$ is the **trace** of A , defined by

$$\text{tr}(A) = \sum_{j=1}^n A_{jj}. \quad (5.42)$$

DEFINITION 5.27 The **Lie algebra of $SU(2^n)$** is

$$\mathfrak{su}(2^n) := \{A \in \mathbb{C}^{2^n \times 2^n} \mid A^\dagger = -A, \text{tr}(A) = 0\}. \quad (5.43)$$

Equivalently,

$$\mathfrak{su}(2^n) = \{-iH \mid H^\dagger = H, \text{tr}(H) = 0\}. \quad (5.44)$$

Thus $\mathfrak{su}(2^n)$ consists exactly of the traceless skew-Hermitian operators.

FACT 5.28 For every $U \in SU(2^n)$, there exists $A \in \mathfrak{su}(2^n)$ such that

$$U = e^A. \quad (5.45)$$

Proof. Since U is unitary, the spectral theorem gives

$$U = V \text{diag}(e^{i\theta_1}, \dots, e^{i\theta_N}) V^\dagger, \quad (5.46)$$

where $N = 2^n$ and $\theta_j \in \mathbb{R}$ for all $j \in [N]$. Since $\det(U) = 1$, we have

$$1 = e^{i(\theta_1 + \dots + \theta_N)}. \quad (5.47)$$

Therefore $\theta_1 + \dots + \theta_N \in 2\pi\mathbb{Z}$. By changing one θ_j by an integer multiple of 2π , we may assume

$$\sum_{j=1}^N \theta_j = 0. \quad (5.48)$$

Now define

$$A := V \text{diag}(i\theta_1, \dots, i\theta_N) V^\dagger. \quad (5.49)$$

Then $A^\dagger = -A$, $\text{tr}(A) = i \sum_j \theta_j = 0$, so $A \in \mathfrak{su}(2^n)$, and

$$e^A = V \text{diag}(e^{i\theta_1}, \dots, e^{i\theta_N}) V^\dagger = U. \quad (5.50)$$

This completes the proof. $\square$

FACT 5.29 The phase-free Pauli strings $P_1 \otimes \cdots \otimes P_n$ form a basis for $\mathbb{C}^{2^n \times 2^n}$. In particular, every Hermitian matrix is a real linear combination of Pauli strings.

FACT 5.30 An operator A lies in $\mathfrak{su}(2^n)$ if and only if there exist real numbers $\alpha_P \in \mathbb{R}$ such that

$$A = -i \sum_{P \in \mathcal{P}_n^\circ} \alpha_P P, \quad (5.51)$$

where $\mathcal{P}_n^\circ := \{P_1 \otimes \cdots \otimes P_n \mid P_j \in \{I, X, Y, Z\}\} \setminus \{I^{\otimes n}\}$.

Proof. First the forward direction. Since each Pauli P is Hermitian, the operator

$$H := \sum_{P \in \mathcal{P}_n^\circ} \alpha_P P \quad (5.52)$$

is Hermitian. Thus $A = -iH$ is skew-Hermitian. As every non-identity Pauli has trace 0,

$$\mathrm{tr}(A) = -i \sum_{P \in \mathcal{P}_n^\circ} \alpha_P \mathrm{tr}(P) = 0. \quad (5.53)$$

Thus $A \in \mathfrak{su}(2^n)$. Conversely, suppose $A \in \mathfrak{su}(2^n)$. Since A is skew-Hermitian, $H := iA$ is Hermitian. Every Hermitian matrix can be written as a real linear combination of Pauli strings, so

$$H = \beta_I I + \sum_{P \in \mathcal{P}_n^\circ} \beta_P P \quad (5.54)$$

for some $\beta_I, \beta_P \in \mathbb{R}$. Taking traces gives

$$0 = \mathrm{tr}(H) = \beta_I \mathrm{tr}(I) + \sum_{P \in \mathcal{P}_n^\circ} \beta_P \mathrm{tr}(P) = 2^n \beta_I. \quad (5.55)$$

Hence $\beta_I = 0$ and

$$H = \sum_{P \in \mathcal{P}_n^\circ} \beta_P P. \quad (5.56)$$

Since $A = -iH$, we get

$$A = -i \sum_{P \in \mathcal{P}_n^\circ} \beta_P P \quad (5.57)$$

as desired. □

THEOREM 5.31 (BAKER-CAMPBELL-HAUSDORFF FORMULA) For sufficiently small $A, B \in \mathfrak{su}(2^n)$,

$$e^A e^B = \exp \left(A + B + \frac{1}{2}[A, B] + \frac{1}{12}[A, [A, B]] + \frac{1}{12}[B, [B, A]] + \cdots \right), \quad (5.58)$$

where $[A, B] = AB - BA$ is the **commutator** [Hal15].

THEOREM 5.32 (TROTTERIZATION) This is the Lie–Trotter formula from [Tro59]. For $A, B \in \mathfrak{su}(2^n)$,

$$e^{A+B} = \lim_{k \rightarrow \infty} (e^{A/k} e^{B/k})^k. \quad (5.59)$$

DEFINITION 5.33 Let $\mathcal{S} \subseteq SU(2^n)$ be a gate set, and let $\mathcal{G} := \overline{\langle \mathcal{S} \rangle}$ be the closed group generated by $\mathcal{S}$. We say that $A \in \mathfrak{su}(2^n)$ is an **available infinitesimal generator** if $e^{tA} \in \mathcal{G}$ for every $t \in \mathbb{R}$.

FACT 5.34 (GENERATING SUMS) If A and B are available infinitesimal generators, then $A + B$ is also available as an infinitesimal generator.

Proof. If e^{tA} and e^{tB} are available for all $t \in \mathbb{R}$, then by Theorem 5.32,

$$e^{t(A+B)} = \lim_{k \rightarrow \infty} (e^{tA/k} e^{tB/k})^k. \quad (5.60)$$

Thus $e^{t(A+B)}$ lies in the closure of the group generated by e^{tA} and e^{tB} . $\square$

FACT 5.35 (GENERATING COMMUTATORS) If A and B are available infinitesimal generators, then $[A, B]$ is also available as an infinitesimal generator.

Proof. Using the BCH formula,

$$e^{\epsilon A} e^{\epsilon B} e^{-\epsilon A} e^{-\epsilon B} = e^{\epsilon^2 [A, B] + O(\epsilon^3)}. \quad (5.61)$$

For $t > 0$, set $\epsilon = t/k^{1/2}$. Then

$$(e^{\epsilon A} e^{\epsilon B} e^{-\epsilon A} e^{-\epsilon B})^k = \left(e^{(t^2/k)[A, B] + O(k^{-3/2})} \right)^k \longrightarrow e^{t^2 [A, B]} \quad (5.62)$$

as $k \rightarrow \infty$. Since $t \in \mathbb{R}$ is arbitrary, this completes the proof. $\square$

This gives $e^{s[A, B]} \in \mathcal{G}$ for every $s \geq 0$. Repeating the same argument with A and B interchanged gives $e^{-s[A, B]} \in \mathcal{G}$. Hence $e^{t[A, B]} \in \mathcal{G}$ for every $t \in \mathbb{R}$.

DEFINITION 5.36 Let $\mathcal{S} \subseteq SU(2^n)$ be a gate set and let $\overline{\langle \mathcal{S} \rangle}$ be the closure of the group generated by $\mathcal{S}$. The **Lie algebra generated by $\mathcal{S}$** is the Lie algebra of $\overline{\langle \mathcal{S} \rangle}$:

$$\mathfrak{su}(\mathcal{S}) := \{A \in \mathfrak{su}(2^n) \mid e^{tA} \in \overline{\langle \mathcal{S} \rangle} \text{ for all } t \in \mathbb{R}\}. \quad (5.63)$$

THEOREM 5.37 Let $\mathcal{S} \subseteq SU(2^n)$. If

$$\mathfrak{su}(\mathcal{S}) = \mathfrak{su}(2^n), \quad (5.64)$$

then $\mathcal{S}$ is approximately universal on n qubits, meaning

$$\overline{\langle \mathcal{S} \rangle} = SU(2^n). \quad (5.65)$$

Proof. Let $G = \overline{\langle \mathcal{S} \rangle}$. By the closed subgroup theorem, G is a matrix Lie subgroup of $SU(2^n)$. Its identity component G° has the same Lie algebra as G . If this Lie algebra is $\mathfrak{su}(2^n)$, then G° is the connected Lie subgroup of $SU(2^n)$ with Lie algebra $\mathfrak{su}(2^n)$. Since $SU(2^n)$ is connected, $G^\circ = SU(2^n)$, and therefore $G = SU(2^n)$. $\square$

DEFINITION 5.38 Let Λ be a n -wire graph gadget and G_Λ its gadget gate. Since global phases are irrelevant, we define its special-unitary representative as follows: Choose any scalar $\omega(\theta) \in \mathbb{U}(1)$ satisfying

$$\omega(\theta)^{2^n} = \det(G_\Lambda(\theta))^{-1}.$$

Define

$$\tilde{G}_\Lambda(\theta) := \omega(\theta)G_\Lambda(\theta) \in SU(2^n).$$

The choice of $\omega(\theta)$ only changes the representative by a global phase.

DEFINITION 5.39 Let Λ be a unitary gadget and G_Λ its gadget gate. Define the closed subgroup generated by the gadget as

$$\mathcal{G}_\Lambda := \overline{\langle \tilde{G}_\Lambda(\theta) \mid \theta \in \mathbb{R}^m \rangle} \subseteq SU(2^n). \quad (5.66)$$

Its Lie algebra is

$$\mathfrak{g}_\Lambda := \mathfrak{su}(\mathcal{G}_\Lambda) \subseteq \mathfrak{su}(2^n). \quad (5.67)$$

LEMMA 5.40 (ZERO-ANGLE CLIFFORD) Let Λ be a unitary gadget whose measurements are all in the XY plane. Then the zero-angle gadget $\tilde{G}_\Lambda(0, \dots, 0)$ is a Clifford operation. In particular,

$$\tilde{G}_\Lambda(0, \dots, 0) \in \mathcal{G}_\Lambda. \quad (5.68)$$

Multiplying by a global phase does not change membership in the Clifford group, so $\tilde{G}_\Lambda(0, \dots, 0)$ is Clifford modulo global phase.

Proof. At angle 0, a XY -plane measurement is an X -basis measurement. The operations are Clifford. The final claim follows because $\tilde{G}_\Lambda(0, \dots, 0)$ is one of the generators used in Definition 5.39. $\square$

LEMMA 5.41 (CLIFFORD INVERSES ARE AVAILABLE) If $C \in \mathcal{G}_\Lambda$ is Clifford, then $C^\dagger \in \mathcal{G}_\Lambda$.

Proof. Modulo global phase, the Clifford group is finite, so there exists $r \geq 1$ and a phase $e^{i\phi}$ such that $C^r = e^{i\phi}I$. Since $C \in SU(2^n)$, this phase is a 2^n -th root of unity. Hence some further power of C is exactly I . Therefore $C^{-1} = C^{s-1} \in \mathcal{G}_\Lambda$ for some $s \geq 1$. $\square$

LEMMA 5.42 (CONJUGATION BY AVAILABLE CLIFFORD GATES) Let $A \in \mathfrak{g}_\Lambda$ and let $C \in \mathcal{G}_\Lambda$ be Clifford with $C^\dagger \in \mathcal{G}_\Lambda$. Then $CAC^\dagger \in \mathfrak{g}_\Lambda$.

Proof. Since $A \in \mathfrak{g}_\Lambda$, we have $e^{tA} \in \mathcal{G}_\Lambda$ for all $t \in \mathbb{R}$. Since $C, C^\dagger \in \mathcal{G}_\Lambda$, $Ce^{tA}C^\dagger \in \mathcal{G}_\Lambda$. But $Ce^{tA}C^\dagger = e^{tCAC^\dagger}$. Thus $e^{tCAC^\dagger} \in \mathcal{G}_\Lambda$ for all $t \in \mathbb{R}$, so $CAC^\dagger \in \mathfrak{g}_\Lambda$. $\square$

FACT 5.43 (ALLOWED LIE-ALGEBRA OPERATIONS) Inside $\mathfrak{g}_\Lambda$, we may use the following closure operations:

$$A, B \in \mathfrak{g}_\Lambda \implies A + B \in \mathfrak{g}_\Lambda, \quad (5.69)$$

$$A, B \in \mathfrak{g}_\Lambda \implies [A, B] \in \mathfrak{g}_\Lambda, \quad (5.70)$$

$$A \in \mathfrak{g}_\Lambda, C, C^\dagger \in \mathcal{G}_\Lambda \implies CAC^\dagger \in \mathfrak{g}_\Lambda. \quad (5.71)$$

The previous fact is precisely the three operations promised at the start of this subsection. Namely, with these operations and some starting operators, the hope is that our gate set will generate the full Lie algebra (Theorem 5.37). The question remains what operators we get to start with.

Proof. The first implication is Fact 5.34. The second implication is Fact 5.35. The third implication is Lemma 5.42. $\square$

FACT 5.44 (INITIAL Z-ROTATIONS) Suppose the gadget Λ has angle parameters allowing the implementation

$$\tilde{G}_\Lambda(2t, 0, \dots, 0) = Ce^{-itZ_1} \quad (5.72)$$

for some Clifford $C \in \mathcal{G}_\Lambda$ and all $t \in \mathbb{R}$. Then $-iZ_1 \in \mathfrak{g}_\Lambda$. More generally, if for each $j \in [n]$ there is a Clifford $C_j \in \mathcal{G}_\Lambda$ such that

$$\tilde{G}_\Lambda(2t, 0, \dots, 0) = e^{-itZ_1}C \quad (5.73)$$

then $-iZ_j \in \mathfrak{g}_\Lambda$ for all $j \in [n]$.

Proof. By Lemma 5.41, $C^\dagger \in \mathcal{G}_\Lambda$. Therefore

$$C^\dagger \tilde{G}_\Lambda(2t, 0, \dots, 0) = e^{-itZ_1} \in \mathcal{G}_\Lambda \quad (5.74)$$

for all $t \in \mathbb{R}$. Hence $-iZ_1 \in \mathfrak{g}_\Lambda$. The same argument applies to each j . $\square$

Note that this does not mean that we have products, e.g. $-iZ_1Z_2$ for “free” too, since $e^{-iZ_1}e^{-iZ_2} = e^{-i(Z_1+Z_2)}$ is the sum rather than the product. Also, no analogous result holds for $-iX_j$ or $-iY_j$, for example, by Lemma 3.9.

THEOREM 5.45 (GADGET UNIVERSALITY VIA PAULI GENERATION) Let Λ be a unitary n -wire gadget. If

$$-iP \in \mathfrak{g}_\Lambda \quad (5.75)$$

for every non-identity tensor-product Pauli $P \in \mathcal{P}_n^\circ$, then Λ is approximately universal on n qubits.

Proof. By Fact 5.30, the elements $-iP$ for non-identity Pauli strings P span $\mathfrak{su}(2^n)$ as a real vector space. Since $\mathfrak{g}_\Lambda$ is closed under real linear combinations by Fact 5.34, $\mathfrak{g}_\Lambda = \mathfrak{su}(2^n)$. Therefore $\mathcal{G}_\Lambda = SU(2^n)$ by Theorem 5.37, so the gate set is approximately universal. $\square$

We are now ready to put everything together into an algorithm.

EXAMPLE 5.46 Consider the diagonal gadget. The relevant Clifford operation is

$$C = \text{CNOT}_{2,0} \text{CZ}_{0,1}(H_0 H_1 H_2). \quad (5.76)$$

Here products are written right-to-left, so $H_0 H_1 H_2$ is applied first, then $\text{CZ}_{0,1}$, then $\text{CNOT}_{2,0}$. Starting from

$$ZII, \quad IZI, \quad IIZ, \quad (5.77)$$

we can generate new operators using $P \mapsto CPC^\dagger$ and closure. The following pseudocode implements exactly this search.

Algorithm 2 3-qubit approximate universality check for the diagonal gadget

```

1:  $C \leftarrow \text{CNOT}_{2,0} \text{CZ}_{0,1}(H_0 H_1 H_2)$ 
2: Queue  $\leftarrow [ZII, IZI, IIZ]$ 
3: Found  $\leftarrow \emptyset$ 
4: while Queue is nonempty do
5:   Pop a Pauli  $P$  from Queue
6:   if  $P \notin$  Found then
7:     Add  $CPC^\dagger$  to Queue
8:     for  $R \in$  Found do
9:       if  $P$  and  $R$  anticommute then
10:        Add the phase-free Pauli underlying  $PR$  to Queue
11:      end if
12:    end for
13:    Add  $P$  to Found
14:   end if
15: end while
16: if Found contains all 63 non-identity three-qubit Paulis then
17:   Conclude  $\mathfrak{g} = \mathfrak{su}(8)$ 
18:   Conclude the gadget is approximately universal
19: end if
```

It turns out that the diagonal gadget fails to generate all 63 non-identity Pauli strings. On the other hand, running the above algorithm for the width-3 gadget succeeds, so the width-3 gadget is at least approximately universal. This result is stronger than it first appears, the notion of which is made precise below.

DEFINITION 5.47 Let $P, Q \subseteq \mathbb{Z}^2$. Say that P and Q are **separated** if the embedding of $P \cup Q$ into CGrid has no edges between them.

LEMMA 5.48 Let $P \subseteq \mathbb{Z}^2$ be finite, and define

$$X := \max_{p,q \in P} |x(p) - x(q)|, \quad Y := \max_{p,q \in P} |y(p) - y(q)|. \quad (5.78)$$

If $t = (A, -B)$ satisfies $A > X$ and $B > Y$, then for every $m \geq 1$, the translates $P, P + t, \dots, P + (m-1)t$ are pairwise separated.

Proof. Let $i < j$, set $d = j - i \geq 1$, and take $p, q \in P$. The displacement from $p + it$ to $q + jt$ is

$$q + jt - (p + it) = q - p + dt. \quad (5.79)$$

Its first coordinate is at least $-X + dA > 0$, while its second coordinate is at most $Y - dB < 0$. Hence the two vertices are incomparable in **CGrid**, so there is no edge between $P + it$ and $P + jt$. $\square$

DEFINITION 5.49 A **k -local layer** on N qubits is a unitary

$$U = \bigotimes_{\alpha=1}^s U_{\alpha}, \quad (5.80)$$

where the supports of the U_{α} 's are disjoint and each U_{α} acts on at most k qubits. A **k -local computation** is a product of k -local layers.

THEOREM 5.50 Let Λ an approximately k -qubit universal tileable graph gadget. Then every k -local layer can be implemented inside one sufficiently large comparability grid. Hence every finite k -local computation can be implemented layer by layer.

Proof. Let $U = \bigotimes_{\alpha=1}^s U_{\alpha}$ be a k -local layer. Since Λ is k -qubit universal, each U_{α} can be approximated to arbitrary accuracy by some finite tiled computation. Choose a computation with the most finite vertices. Put all vertices of one such computation inside a finite point set P . Choose $t = (A, -B)$ with

$$A > \max_{p,q \in P} |x(p) - x(q)|, \quad B > \max_{p,q \in P} |y(p) - y(q)|. \quad (5.81)$$

By Lemma 5.48, the translated regions $P, P + t, P + 2t, \dots$ have no edges between distinct copies. Thus the computations for the U_{α} 's may be placed independently along an anti-diagonal of one large **CGrid**, and the resulting operation is exactly

$$\bigotimes_{\alpha=1}^s U_{\alpha}. \quad (5.82)$$

Repeating this construction for each layer implements any finite product of k -local layers. $\square$

Thus, we may say that the width 3 gadget is approximately universal for 3-local layers.

6 Conclusions

In Section 5, we introduced a series of algorithms for developing and testing finite graph gadgets. Given a tileable graph gadget Λ , we associated to it a parametrized gate G_Λ and tested the approximate universality of the resulting gate family. In particular, the diagonal gadget fails the 3-qubit approximate universality test, while the width 3 gadget passes.

In this section, we present a different kind of construction. Instead of searching for one finite tile, we define infinite families of gadgets $\hat{\Lambda} = (\Lambda_k)_{k \geq 1}$ where Λ_k is intended to act on k wires.

DEFINITION 6.1 An **iterative graph gadget** is a sequence

$$\hat{\Lambda} = (\Lambda_k)_{k \geq 1}, \quad (6.1)$$

where each $\Lambda_k = (H_k, L_k, R_k)$ is a k -wire graph gadget. We write

$$L_k = (\ell_1^{(k)}, \dots, \ell_k^{(k)}), \quad R_k = (r_1^{(k)}, \dots, r_k^{(k)}) \quad (6.2)$$

for its ordered left and right boundaries.

DEFINITION 6.2 The **middle-all-connected iterative gadget**

$$\hat{\Lambda}^{\text{mid}} = (\Lambda_k^{\text{mid}})_{k \geq 1} \quad (6.3)$$

is defined inductively as follows. For $k = 1$, let Λ_1^{mid} be the one-edge line gadget. For $k = 2$, let Λ_2^{mid} be the $\text{Grid}_{2 \times 2}$ gadget.

Suppose $\Lambda_k^{\text{mid}} = (H_k, L_k, R_k)$ has been defined, with $L_k = (\ell_1, \dots, \ell_k)$, and $R_k = (r_1, \dots, r_k)$. To construct $\Lambda_{k+1}^{\text{mid}}$, take two copies H_k^{left} and H_k^{right} , and glue the right boundary of the left copy to the left boundary of the right copy:

$$r_j^{\text{left}} = \ell_j^{\text{right}}, \quad \text{for every } j \in [k]. \quad (6.4)$$

Let Q_k denote the result.

Now add three new vertices a_k, b_k, c_k . The vertex a_k is adjacent to b_k and to every left input of H_k^{left} . The vertex c_k is adjacent to b_k and to every right output of H_k^{right} . The vertex c_k is adjacent to every vertex of Q_k , and also to a_k and b_k . No other new edges are added. Equivalently,

$$N(a_k) = \{b_k\} \cup L_k^{\text{left}}, \quad (6.5)$$

$$N(b_k) = V(Q_k) \cup \{a_k, c_k\}, \quad (6.6)$$

$$N(c_k) = \{b_k\} \cup R_k^{\text{right}}. \quad (6.7)$$

The new boundaries are

$$L_{k+1} := (a_k, \ell_1^{\text{left}}, \dots, \ell_k^{\text{left}}), \quad (6.8)$$

$$R_{k+1} := (c_k, r_1^{\text{right}}, \dots, r_k^{\text{right}}). \quad (6.9)$$

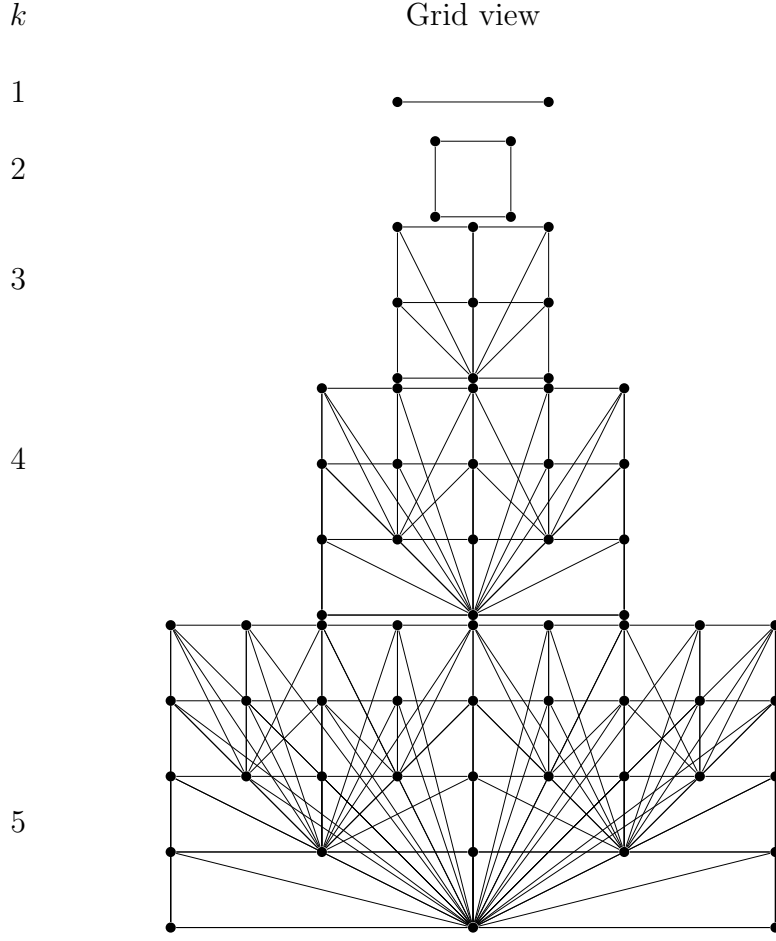


Figure 16: Grid views of the first five middle-all-connected iterative gadgets Λ_k^{mid} .

LEMMA 6.3 Let Q be a finite induced subgraph of some comparability grid, equipped with a fixed coordinate embedding $V(Q) \subseteq \mathbb{Z}^2$, and let $L = (\ell_1, \dots, \ell_k)$, $R = (r_1, \dots, r_k)$ be ordered boundary sets. Suppose Q is embedded so that there exist real numbers $\gamma < \alpha$ with

$$\max_{p \in V(Q) \setminus L} y(p) < \alpha < \min_j y(\ell_j), \quad \max_j y(r_j) < \gamma < \min_{p \in V(Q) \setminus R} y(p).$$

Then one may add three vertices a, b, c so that the resulting graph remains an induced subgraph of some CGrid, and

$$N(a) = \{b\} \cup L, \tag{6.10}$$

$$N(b) = V(Q) \cup \{a, c\}, \tag{6.11}$$

$$N(c) = \{b\} \cup R. \tag{6.12}$$

Proof. Let $P \subseteq \mathbb{Z}^2$ be the point set realizing Q . Choose

$$x_- < \min_{p \in P} x(p), \quad x_+ > \max_{p \in P} x(p). \tag{6.13}$$

By the separation hypotheses, choose real numbers α, γ such that

$$\max_{p \in V(Q) \setminus L} y(p) < \alpha < \min_{j \in [k]} y(\ell_j), \quad (6.14)$$

$$\max_{j \in [k]} y(r_j) < \gamma < \min_{p \in V(Q) \setminus R} y(p), \quad (6.15)$$

and choose them with $\alpha > \gamma$. Finally choose

$$\beta < \min\{\gamma, \min_{p \in P} y(p)\}. \quad (6.16)$$

Place

$$a := (x_-, \alpha), \quad (6.17)$$

$$b := (x_- - 1, \beta), \quad (6.18)$$

$$c := (x_+, \gamma). \quad (6.19)$$

Since b is below-left of every point in P , and also below-left of a and c , it is adjacent to every vertex of Q and to both a and c . Now a lies to the left of every point of P . Hence a is adjacent to a point $p \in P$ exactly when $\alpha \leq y(p)$. By construction, this holds exactly for $p \in L$. Thus

$$N(a) = \{b\} \cup L. \quad (6.20)$$

Similarly, c lies to the right of every point of P . Hence c is adjacent to $p \in P$ exactly when $y(p) \leq \gamma$. By construction, this holds exactly for $p \in R$. Thus

$$N(c) = \{b\} \cup R. \quad (6.21)$$

Finally, since $x(a) < x(c)$ but $y(a) > y(c)$, the vertices a and c are incomparable. Therefore no other new edges are introduced, so the extension is induced in **CGrid**. $\square$

THEOREM 6.4 For every $k \geq 1$, the middle-all-connected gadget $\Lambda_k^{\text{mid}} = (H_k, L_k, R_k)$ admits an induced embedding into a comparability grid in which L_k is separated above the rest of the graph and R_k is separated below the rest of the graph.

Proof. We prove by induction on k . The cases $k = 1$ and $k = 2$ are realized directly by the line gadget and the 2×2 grid gadget.

Assume Λ_k^{mid} has such a separated embedding. Take two copies, translate them far apart if necessary, and glue the right boundary of the left copy to the left boundary of the right copy as in the definition. The resulting graph Q_k again satisfies the separation hypotheses. Lemma 6.3 then adds a_k, b_k, c_k without adding additional edges and produces the required separated embedding for $\Lambda_{k+1}^{\text{mid}}$. $\square$

CONJECTURE 6.5 $\hat{\Lambda}^{\text{mid}}$ is approximately universal for MBQC. That is, for every $k \in \mathbb{N}$, Λ_k^{mid} is approximately k -qubit universal.

Evidence. Running the algorithm in Section 5.4 on Λ_k^{mid} for $k \in [5]$ yields that Λ_k^{mid} is approximately universal for k qubits for every $k \in [5]$. $\square$

We now take a moment to discuss this result. First, note that the number of vertices in Λ_k^{mid} grows exponentially in k . This means that Theorem 4.10 does not eliminate the possibility that the conjecture is true. On the other hand, a proof for all $k \in \mathbb{N}$ is tricky primarily for two reasons. First, when constructing $\Lambda_{k+1}^{\text{mid}}$, we introduce many new edges from b to every other vertex. This almost certainly affects the computation of H_k^{left} and H_k^{right} . Second, assuming we have shown that Λ_k^{mid} is approximately universal, this almost certainly does not imply that a single $G_{\Lambda_k^{\text{mid}}}$ is k -qubit approximate universal. Indeed, we may need to apply $G_{\Lambda_k^{\text{mid}}}$ many times to attain this. Both of these complications make a proof of the conjecture, and similar ones to it, challenging.

We now conclude this section and the thesis by presenting some future directions.

1. An immediate next step is to develop an algorithmic pipeline that incorporates local complementation into every part of the construction. The hope is that local complementation may introduce the additional structure and tools needed to prove conjectures like the one above.
2. Another future direction is to better characterize what makes families of graph states universal. In our particular case, a potential next step is to understand more precisely why the middle-all-connected construction works up to approximate 5-qubit universality, and why other constructions don't.
3. We may attempt a new approach to the problem. Suppose instead that we have a random variable that gives us induced subgraphs of **CGrid** that themselves contain some **Grid** as a subgraph. Can probabilistic methods be used to show universality this way?
4. We have seen how the theory of vertex-minors and the rank-width measure are very powerful concepts for characterizing the universality of graph state families in general. What other measures are available, and can they be applied to the comparability grid?

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A Appendix: Background on Quantum Computing

A.1 Qubits

DEFINITION A.1 A single **qubit** $|\phi\rangle$ is a state

$$|\phi\rangle = \alpha \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \end{bmatrix} \in \mathbb{C}^2, \quad (\text{A.1})$$

where $|\alpha|^2 + |\beta|^2 = 1$ and $\alpha, \beta \in \mathbb{C}$. We say $|\phi\rangle$ is in a **superposition** of $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Thus a single qubit is a length 2 vector in the Hilbert space $\mathbb{C}^2$ with ℓ_2 norm 1.

DEFINITION A.2 (BRAKET NOTATION) Define “kets” $|0\rangle := \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle := \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, both in $\mathbb{C}^2$. For any $|\phi\rangle$, let “bra” $\langle\phi| := |\phi\rangle^\dagger = (|\phi\rangle^*)^T$ be its conjugate transpose. Thus the bracket denotes the inner product:

$$\langle\phi|\psi\rangle := \langle\phi| \cdot |\psi\rangle. \quad (\text{A.2})$$

Now the ℓ_2 norm of $|\phi\rangle$ can be written as $\| |\phi\rangle \|_2 = \sqrt{\langle\phi|\phi\rangle}$.

DEFINITION A.3 Let $A \in \mathbb{C}^{m \times n}$ and $B \in \mathbb{C}^{o \times p}$. For our purposes, it suffices to define the **tensor product**

$$A \otimes B := \begin{bmatrix} a_{1,1}B & \cdots & a_{1,n}B \\ \vdots & \ddots & \vdots \\ a_{m,1}B & \cdots & a_{m,n}B \end{bmatrix}. \quad (\text{A.3})$$

where $a_{i,j}$ denotes the entry of A in row i and column j .

DEFINITION A.4 Let $x = \overline{x_1 x_2 \cdots x_d} \in \{0, 1\}^d$, i.e. a binary bit-string. Define

$$|x\rangle := \bigotimes_{i=1}^d |x_i\rangle. \quad (\text{A.4})$$

For example, $|00\rangle = |0\rangle \otimes |0\rangle = [1 \ 0 \ 0 \ 0]^T$. We may also replace x by its integer evaluation as a binary number, by abuse of notation. For example,

$$|3\rangle = |11\rangle = [0 \ 0 \ 0 \ 1]^T \quad (\text{A.5})$$

but we may also have

$$|3\rangle = |011\rangle = [0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0]^T \quad (\text{A.6})$$

Which binary representation we choose should be clear from context. For example, this gives us two equivalent definitions of a q -qubit state

$$|\varphi\rangle = \sum_{x \in \{0,1\}^d} \alpha_x |x\rangle = \sum_{x=0}^{2^q-1} \alpha_x |x\rangle \in \mathbb{C}^{2^q} \quad (\text{A.7})$$

where $\alpha_x \in \mathbb{C}$ and $\sum_{x \in \{0,1\}^d} |\alpha_x|^2 = 1$. As the number of dimensions increases exponentially with respect to the number of qubits, bracket notation will become increasingly concise compared to traditional vector notation.

DEFINITION A.5 For brevity, we will write $|\phi\rangle|\psi\rangle = |\phi\rangle \otimes |\psi\rangle$, so adjacent kets imply $\otimes$. However, note that $|\phi\psi\rangle \neq |\phi\rangle|\psi\rangle$ in general. Also, let

$$A^{\otimes n} := \bigotimes_{i=1}^n A. \quad (\text{A.8})$$

DEFINITION A.6 Vectors $|v_1\rangle, \dots, |v_n\rangle \in \mathbb{C}^n$ are an orthonormal basis if

1. $\| |v_i\rangle \|_2 = 1$ for every $i \in [n]$ and
2. $|v_i\rangle \perp |v_j\rangle$ (equivalently, $\langle v_i | v_j \rangle = 0$) for every $i, j \in [n]$ with $i \neq j$.

Henceforth, all bases are orthonormal unless explicitly said otherwise.

DEFINITION A.7 (MEASUREMENT) The **Born Rule** gives outcome probabilities for quantum measurements: An n -qubit state $|\phi\rangle$ can be measured under any orthonormal basis $\{|v_i\rangle\}_{i \in [2^n]}$ of $\mathbb{C}^{2^n}$. When measured, the outcome i occurs with probability $|\langle v_i | \phi \rangle|^2$. After seeing outcome i , the state collapses to $|v_i\rangle$. This process is irreversible.

DEFINITION A.8 One can also view measurement as projection. This is useful for **partial measurement**, where only some qubits in a state are measured. Let the n -qubit space $\mathbb{C}^{2^n} = S_1 \otimes S_2$, where $S_1 \cong (\mathbb{C}^2)^{\otimes k}$. Let $\{|s_i\rangle\}_{i \in [2^k]}$ be an orthonormal basis for S_1 . On outcome $i \in [2^k]$, define the **projection matrix** on S_1 as

$$P_i := |s_i\rangle \langle s_i|, \quad \text{such that} \quad P_i = P_i^\dagger, \quad P_i^2 = P_i. \quad (\text{A.9})$$

Define the **partial measurement** projector on S_1 by $M_i = P_i \otimes I \in \mathbb{C}^{2^n \times 2^n}$. For a state $|\phi\rangle \in \mathbb{C}^{2^n}$,

$$\Pr(\text{Outcome } i \mid |\phi\rangle) = \langle \phi | M_i | \phi \rangle = \|M_i | \phi \rangle\|^2. \quad (\text{A.10})$$

Conditioned on outcome i , the post-measurement state becomes

$$|\phi_i\rangle = \frac{M_i | \phi \rangle}{\sqrt{\langle \phi | M_i | \phi \rangle}}. \quad (\text{A.11})$$

EXAMPLE A.9 Consider the states

$$|+\rangle := \frac{1}{\sqrt{2}} |0\rangle + \frac{1}{\sqrt{2}} |1\rangle, \quad |-\rangle := \frac{1}{\sqrt{2}} |0\rangle - \frac{1}{\sqrt{2}} |1\rangle. \quad (\text{A.12})$$

Measuring either in the **standard basis** $\{|0\rangle, |1\rangle\}$ gives $|0\rangle$ with $\frac{1}{2}$ probability since

$$|\langle 0 | + \rangle|^2 = \left| \frac{1}{\sqrt{2}} \cdot 1 + \frac{1}{\sqrt{2}} \cdot 0 \right|^2 = \frac{1}{2} \quad (\text{A.13})$$

and similarly for $|1\rangle$. The squaring makes $|-\rangle$ proceed similarly as well.

DEFINITION A.10 A two-qubit state $|\phi\rangle$ is **entangled** if it cannot be written as the tensor product of single qubits. That is, there does not exist single-qubit states $|\psi\rangle, |\varphi\rangle \in \mathbb{C}^2$ such that $|\phi\rangle = |\psi\rangle \otimes |\varphi\rangle$. If a state is not entangled, it is called **separable**. A n -qubit state $|\psi\rangle$ is entangled (or *contains* entanglement) if it is not fully separable.

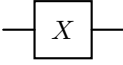
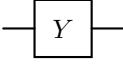
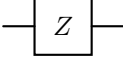
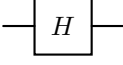
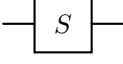
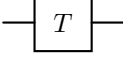
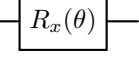
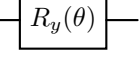
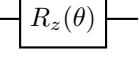
Gate	Circuit	Matrix
X		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Y		$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Z		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
H		$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
S		$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
T		$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
$R_x(\theta)$		$\begin{bmatrix} \cos(\theta/2) & -i \sin(\theta/2) \\ -i \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$
$R_y(\theta)$		$\begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$
$R_z(\theta)$		$\begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}$

Table 2: Common single qubit quantum gates

A.2 Quantum Gates

DEFINITION A.11 A one-qubit quantum gate U is a matrix $U \in \mathbb{C}^{2 \times 2}$ with:

1. $U^\dagger U = U U^\dagger = I$;
2. equivalently, $U = |\phi_0\rangle \langle 0| + |\phi_1\rangle \langle 1|$, where $\{|\phi_0\rangle, |\phi_1\rangle\}$ forms an orthonormal basis.

Common single-qubit gates are given in Table 2.

THEOREM A.12 (EULER DECOMPOSITION) Every one-qubit unitary U can be written as

$$U = e^{i\alpha} R_Z(\beta) R_X(\gamma) R_Z(\delta), \quad (\text{A.14})$$

where $\alpha, \beta, \delta \in [0, 2\pi)$ and $\gamma \in [0, \pi]$.

Proof. This is a standard proof, seen for example in Section 4.2 of [NC10]. Since U is unitary, $|\det U| = 1$. Choose $\alpha \in [0, 2\pi)$ such that $\det U = e^{2i\alpha}$. Let $V := e^{-i\alpha}U$ satisfy $\det V = 1$ while also being unitary. Now let

$$V = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad (\text{A.15})$$

Since the columns of V are orthonormal, the second column must be a unit vector orthogonal to the first. Thus

$$\begin{bmatrix} b \\ d \end{bmatrix} = \lambda \begin{bmatrix} -\bar{c} \\ \bar{a} \end{bmatrix} \quad (\text{A.16})$$

for some $|\lambda| = 1$. Hence

$$V = \begin{bmatrix} a & -\lambda\bar{c} \\ c & \lambda\bar{a} \end{bmatrix}. \quad (\text{A.17})$$

Taking determinants gives

$$1 = \det V = \lambda|a|^2 + \lambda|c|^2 \quad (\text{A.18})$$

$$= \lambda(|a|^2 + |c|^2) \quad (\text{A.19})$$

$$= \lambda. \quad (\text{A.20})$$

This also gives $|a|^2 + |c|^2 = 1$. Choose $\gamma, \eta, \mu \in \mathbb{R}$ such that

$$a = e^{i\eta} \cos(\gamma/2), \quad (\text{A.21})$$

$$c = e^{i\mu} \sin(\gamma/2). \quad (\text{A.22})$$

Then

$$V = \begin{bmatrix} e^{i\eta} \cos(\gamma/2) & -e^{-i\mu} \sin(\gamma/2) \\ e^{i\mu} \sin(\gamma/2) & e^{-i\eta} \cos(\gamma/2) \end{bmatrix}. \quad (\text{A.23})$$

On the other hand, matrix multiplication gives

$$R_Z(\beta)R_X(\gamma)R_Z(\delta) = \begin{bmatrix} e^{-i(\beta+\delta)/2} \cos(\gamma/2) & -ie^{-i(\beta-\delta)/2} \sin(\gamma/2) \\ -ie^{i(\beta-\delta)/2} \sin(\gamma/2) & e^{i(\beta+\delta)/2} \cos(\gamma/2) \end{bmatrix}. \quad (\text{A.24})$$

Thus it is enough to choose β, δ so that

$$e^{i\eta} = e^{-i(\beta+\delta)/2}, \quad (\text{A.25})$$

$$e^{i\mu} = -ie^{i(\beta-\delta)/2}. \quad (\text{A.26})$$

Equivalently, modulo 2π , solving gives

$$\beta = \mu - \eta + \frac{\pi}{2}, \quad (\text{A.27})$$

$$\delta = -\mu - \eta - \frac{\pi}{2}. \quad (\text{A.28})$$

Therefore $V = R_Z(\beta)R_X(\gamma)R_Z(\delta)$. Since $U = e^{i\alpha}V$, we get

$$U = e^{i\alpha}R_Z(\beta)R_X(\gamma)R_Z(\delta) \quad (\text{A.29})$$

as desired. $\square$

FACT A.13 For matrices A, B, C, D over $\mathbb{C}$ of compatible dimensions, scalar $\lambda \in \mathbb{C}$, and vectors $|\phi\rangle$ and $|\psi\rangle$ over $\mathbb{C}$, the tensor product satisfies:

1. Compatability with matrix multiplication: $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$, so $(A \otimes B)(|\phi\rangle \otimes |\psi\rangle) = (A|\phi\rangle) \otimes (B|\psi\rangle)$.
2. Scalars commute: $\lambda(A \otimes B) = (\lambda A) \otimes B = A \otimes (\lambda B)$.
3. Associativity: $(A \otimes B) \otimes C = A \otimes (B \otimes C)$.
4. Unitary-preserving: If A, B are unitary matrices, $A \otimes B$ is unitary.

DEFINITION A.14 An n -**qubit gate** is a unitary matrix in $\mathbb{C}^{2^n \times 2^n}$. If U_i is an n_i -qubit gate for $i = 1, \dots, k$, and $\sum_i n_i = n$, then

$$U_1 \otimes \dots \otimes U_k \tag{A.30}$$

is an n -qubit gate by Fact A.13. Special named multi-qubit gates are given in Table 3.

DEFINITION A.15 Many of the gates in Table 3 are **controlled** gates. Informally, controlled gates such as CNOT only apply their target operation when all controls $\text{---}\bullet\text{---}$ are in the state $|1\rangle$. Formally, for any n -qubit gate U , the singly controlled gate, denoted C- U , is defined by

$$\text{C-}U := |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U. \tag{A.31}$$

This definition generalizes naturally to multiple controls.

By the above definition, CNOT is just C- X , and the Toffoli gate is CC- X . Note that we may suppress the hyphen when the meaning is clear.

The gate C- Z is special because either qubit can be interpreted as the control: it only applies a phase when both qubits are in the state $|1\rangle$. To see this, note that

$$I = |0\rangle\langle 0| + |1\rangle\langle 1|, \quad Z = I - 2|1\rangle\langle 1|. \tag{A.32}$$

Therefore,

$$\text{C-}Z = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes (I - 2|1\rangle\langle 1|) \tag{A.33}$$

$$= I \otimes I - 2|11\rangle\langle 11|. \tag{A.34}$$

Thus C- Z acts as the identity on every computational basis state except $|11\rangle$, where it applies a phase of -1 .

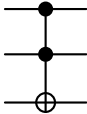
Gate	Circuit	Matrix
CNOT		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
CCNOT or Toffoli		$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$
SWAP		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
C-Z		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$

Table 3: Common multi-qubit quantum gates

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